

Article

Decision-Making Model for Life Cycle Management of Aircraft Components

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Abstract: This paper presents a novel decision-making framework for the life cycle management of aircraft components, integrating advanced data analytics, artificial intelligence, and predictive maintenance strategies. The proposed model addresses the challenges of balancing safety, reliability, and cost-effectiveness in aircraft maintenance. By using real-time health monitoring systems, failure probability models, and economic analysis, the framework enables more informed and dynamic maintenance strategies. The model incorporates a comprehensive approach that combines reliability assessment, economic analysis, and continuous re-evaluation to optimize maintenance, replacement, and life extension decisions. The optimization method on the base of genetic algorithm (GA) is employed to minimize total life cycle costs while maintaining component reliability within acceptable thresholds. The framework's effectiveness is demonstrated through case studies on three distinct aircraft components: mechanical, avionics, and engine. These studies showcase the model's versatility in handling different failure patterns and maintenance requirements. This study introduces a data-driven decision-making framework for optimizing the life cycle management of aircraft components, focusing on reliability, cost-effectiveness, and safety. To achieve optimal maintenance scheduling and resource allocation, a GA is employed, allowing for an effective exploration of complex solution spaces and enabling dynamic decision-making based on real-time data inputs. The GA-based optimization approach minimizes total life cycle costs while maintaining component reliability, with the framework's effectiveness demonstrated through case studies on key aircraft components. Key findings from the case study demonstrate significant cost reductions through optimization, with mechanical components showing a 10% more reduction in total life cycle costs, avionics components achieving a 14% more cost reduction, and engine components demonstrating a 7% more decrease in total costs. The research also presents an optimized dynamic maintenance schedule that adapts to real-time component health data, extending component lifespans and reducing unexpected failures. The framework effectively addresses key industry challenges such as no fault found events while minimizing unexpected failures and enhancing the overall reliability and safety of aircraft maintenance practices. Sensitivity analysis further demonstrates the model's robustness, showing stable performance under varying failure rates, maintenance costs, and degradation rates. The study contributes a scalable approach to predictive maintenance, balancing safety, cost, and resource allocation in dynamic operational environments.

Keywords: aircraft maintenance; life cycle cost optimization; predictive maintenance; condition-based maintenance; genetic algorithm; health index; reliability assessment; economic analysis; dynamic maintenance scheduling; aviation component management

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1. Introduction

1.1. Background and Motivation

The aviation industry stands at a critical juncture where the confluence of increasing operational demands, stringent safety regulations, and economic pressures necessitates a

paradigm shift in aircraft maintenance and component lifecycle management [1]. Modern aircrafts, with their complex systems and sophisticated components, generate vast amounts of data throughout their operational life. These data hold the potential to revolutionize the maintenance, repair, and overhaul (MRO) processes, moving from reactive and scheduled maintenance to predictive and prescriptive strategies.

Traditional approaches to aircraft component lifecycle management, while effective, are often based on conservative estimates and fixed schedules. A typical maintenance procedure for an aircraft component follows a structured process. Initially, the component arrives at the maintenance facility, where it undergoes a visual inspection. Following this, the component is subjected to a series of tests, including electrical and mechanical assessments, to identify any faults or deviations from normal operation. If issues are detected, the component proceeds to the repair stage, where necessary fixes are implemented. Post-repair, the component undergoes another round of testing to ensure the effectiveness of the repairs. If no issues are found during the initial testing or after repairs, the component moves to the final inspection stage. Here, it is thoroughly examined to confirm its airworthiness. Once approved, the component is certified as serviceable and prepared for return to operation. Throughout this process, detailed documentation is maintained, recording all findings, repairs, and test results, ensuring a comprehensive history of the component's maintenance journey.

These methods, rooted in decades of experience and regulatory compliance, have served the industry well in maintaining high safety standards. However, they frequently result in premature component replacements, unnecessary downtime, and suboptimal resource allocation. The challenge lies in using the wealth of available data to make more informed, timely, and cost-effective decisions without compromising safety.

The current MRO process of components represents a standardized procedure regulated by relevant legislation and procedures outlined in the maintenance manuals for the components.

Within the MRO process of components, problems of the different nature are periodically encountered:

- No fault found (NFF)—a state in which all types of tests show that the component is working properly, and no deviations are observed. However, when installing such a component on an aircraft, the aircraft systems detect an error, and the component once again goes for repair. Such cases have a cyclical nature and, due to the peculiarities and contracts between organizations, can circulate without undergoing repair for as long as necessary. The NFF condition is especially characteristic of avionics components, and the NFF rate can reach from 20–30% [2–4] to 50% [5,6] and more [7]. This results in enormous losses for airlines due to re-routing aircraft to maintenance, additional unscheduled aircraft downtime, and the need for additional purchase of components.
- Replacement part availability when, during maintenance, defects and conditions of components are identified that require replacement of these components. As a rule, these components are not manufactured for sale but only in the quantity necessary to produce and assemble the required number of components.
- Lack of analytics and feedback when organization responsible for the MRO of components often does not have appropriate processes in place to perform analytics and develop modifications to component parts to improve their reliability. Moreover, such organizations typically do not inform the original manufacturers of the components about which component parts fail or offer recommendations for improving the reliability of these component parts. At the same time, the component manufacturers themselves do not have the resources or authority to require MRO organizations and airlines to provide information on such defects.

These challenges highlight the need for improved communication, data analysis, and lifecycle management in aircraft component maintenance and repair.

This research aims to provide a robust, data-driven approach to aircraft component lifecycle management, potentially transforming how the aviation industry approaches maintenance and reliability.

1.2. Related Works

Structural life and reliability management are vital for maintaining an aircraft's flight safety [8]. Over the past few decades, the concept of "aircraft fleet management" has been widely adopted. This approach entails managing all aircraft in the fleet according to a uniform standard, based on the evaluated reliable life determined through full-scale fatigue tests (e.g., flight hours, flight cycles, and calendar life at a specified reliability level). Many aircraft types still employ this fleet-wide life management strategy. However, actual service loads experienced by aircrafts can differ significantly from test or design loads, and each aircraft within a fleet can be subjected to varying service loads [9,10]. As a result, individual aircrafts have different service lives. Aircrafts exposed to more severe loads may have a shorter service life, making uniform fleet management based on general life indexes potentially dangerous. On the other hand, aircrafts with lighter loads could have their life potential underestimated, leading to economic inefficiencies [11].

To address both safety and economic concerns, the concept of individual life monitoring (or individual aircraft tracking) has gained considerable interest in both industry and academia, especially for aging and new aircrafts alike [12]. The primary procedure for individual aircraft life monitoring includes load monitoring, damage assessment, and life prediction and management [13]. Load monitoring, which is fundamental to this process, is typically conducted through flight parameter monitoring systems and strain gauges positioned on critical parts of the aircraft. Techniques such as finite element analysis, multiple linear regression, and artificial neural networks are used to assess loads accurately [14–17], establishing a strong foundation for individual aircraft life monitoring.

Once the service loads are obtained, the focus shifts to defining and evaluating damage caused by these loads. The core of individual aircraft life monitoring involves damage definition and assessment [11]. Common damage indexes used today include the fatigue index, fatigue damage index, fatigue life expended, fatigue life expended index, and crack severity index [9,18–21]. These damage indexes are categorized into two main approaches. The first defines damage by evaluating fatigue life under different load conditions and applies Miner's linear cumulative damage theory to assess damage accumulation or life consumption [11]. The second approach focuses on damage resulting from crack growth under varying loads, with damage tolerance theory forming the technical basis [19].

Aircraft health monitoring systems (AHMSs) focus on continuously monitoring maintenance-critical components, presenting new opportunities for designing lighter, safer, and more efficient aircraft [22]. AHMSs aim to eliminate or reduce the need for traditional design accommodations, such as composite knockdown factors and strict scheduled inspections, leading to more cost-effective maintenance strategies [23]. Over the past few decades, various approaches and technologies have been developed to support AHMSs, notably the use of permanently installed sensors on or embedded within the airframe. These sensors provide both global and local feedback on structural health, enabling real-time, on-demand measurements that allow for trend monitoring in system behavior [24]. By collecting time or frequency domain data throughout the aircraft's life, AHMSs can identify anomalies, track their location and severity, and estimate the remaining useful life of monitored components, which may extend inspection intervals [25].

Researchers have investigated several diagnostic techniques for aircraft structures, including ultrasonic guided waves [26,27], guided electromagnetic waves [28], electromechanical impedance [29], and vibration response [30]. AHMSs can be passive, activated by the aircraft boundary layer [31], or active, powered by the system itself [32]. In either case, a damage feature sensitive to defects in the observed structure is used. The system correlates measurements from sensor clusters to detect and assess damage. Various monitoring methods are designed to evaluate specific damage types, and multi-sensor approaches have

gained considerable attention [33]. Integrating transducers onboard is critical for effective system monitoring, and this approach is also applied to aircraft systems like actuators, where existing or new transducers are used to assess system health [34].

Understanding the reliability and affordability of AHMSs is essential for their successful integration. Currently, the lack of reliability and cost-benefit assessments limits the industrial deployment of on-condition maintenance systems. The reliability of an AHMS depends on the specific approach used and is increasingly addressed through collaborations between AHMS experts and reliability specialists [35]. These collaborations aim to determine the minimum detectable damage size throughout an aircraft's life, helping to guide AHMS implementation. On the other hand, conducting a cost-benefit analysis is more complex, as it requires estimating both the benefits (e.g., maintenance savings and optimized structural design) and the costs (e.g., system installation and added weight). A multidisciplinary analysis that considers health management, design, and performance parameters, independent of the specific AHMS technology, is necessary for determining the profitability and efficiency of integrating an AHMS into the aircraft [35].

One of the significant challenges in modern aircraft design is the use of novel materials, such as composites, in load-bearing structures. Composites offer performance improvements but come with critical challenges due to their anisotropic properties. For instance, their stacking characteristics and lower strength in the normal direction make them susceptible to failure under dynamic loads, leading to delamination caused by high interlaminar shear stress [36–38]. Low-velocity impacts can randomly cause failures that, if undetected, could threaten the safety of an aircraft mission. These safety-critical aspects require appropriate design approaches, often driven by damage tolerance methodologies that assume undetectable damage, raising concerns about the extent of weight savings achieved through effective damage detection.

The safety-by-design philosophy of the aviation industry, established in the 1980s, incorporates the damage tolerance approach, which combines inspection with structural design concepts to ensure safety while adhering to inspection procedures [39–41]. However, this approach does not fully exploit material capabilities due to the sensitivity and costs associated with inspections. This results in higher maintenance costs and increases direct operating costs from excessive safety margins and critical operations. AHMSs offer significant potential benefits by continuously monitoring aircraft for damage and allowing for more flexible and faster maintenance schedules, improved safety, and more relaxed design constraints.

Fioriti et al. [42] found that effective prognostics could increase aircraft availability and profitability for airlines. However, utilizing AHMS-derived information for condition-based maintenance remains critical. Current discussions on integrating real-time monitoring with relative cost-benefit analysis are limited and often unrealistic, relying heavily on wireless connections, onboard sensors, and ground-based computing stations [43,44]. Addressing technological gaps in AHMS operation could reduce the need for visual and non-destructive inspections, which contribute to maintenance downtime, and streamline health management tasks [45].

Modern aircraft operator maintenance programs are primarily based on regulatory and industry guidelines rooted in reliability-centered maintenance processes [46] and the latest Maintenance Steering Group (MSG-3) methodology [47], which evolved from the initial approach developed for the Boeing 747–100 in 1968. The original methodology, MSG-1, focused on preventive maintenance programs designed to ensure operational safety and uncover hidden functional failures through interval-driven, hard-time limited, and on-condition inspection tasks. The subsequent update, MSG-2, introduced in 1979, expanded the processes by incorporating condition-based maintenance practices that allowed for monitoring aircraft systems at the component level. However, this bottom-up approach led to an increased economic burden due to a rise in the number of tasks.

The latest iteration, MSG-3, adopts a top-down approach, focusing on the impact of system failures on operational costs and safety if scheduled maintenance is not performed.

This shift in methodology has reduced the number of maintenance tasks, thereby increasing aircraft availability and optimizing maintenance efficiency [48].

In contrast, military aircrafts are typically designed using the damage tolerance analysis principle, a deterministic fail-safe approach for aircraft sustainment [49,50]. Military aircrafts endure more structurally damaging flight hours compared to commercial passenger aircrafts as they experience higher normal load factors and more frequent occurrences of variable operational conditions [51]. However, the fail-safe approach used in military aircraft maintenance does not account for the individual conditions of each aircraft. As a result, maintenance programs tend to rely on conservative scatter and safety factors, leading to overly cautious life estimates and inspection intervals, which reduce efficiency.

To address these inefficiencies, the implementation of condition-based maintenance, supported by structural health and prognostics management technologies, has the potential to significantly reduce inspection costs, extend the lifespan of aircraft structures, and improve overall aircraft availability [52]. This transition to condition-based maintenance, with the aid of advanced monitoring technologies, represents a major advancement in modern maintenance practices for both commercial and military aviation.

1.3. Research Gap, Contributions, and Paper Structure

Although predictive maintenance techniques have been explored in aviation, there is limited research on how to effectively incorporate these techniques into a dynamic, adaptive decision-making process for component lifecycle management. While previous studies have applied various optimization techniques and data-driven approaches to component life cycle management, they often fall short in integrating real-time health assessment with a fully adaptive, condition-based optimization framework. Many existing models rely on fixed schedules or use single metrics for health assessment, limiting their ability to respond dynamically to real-time component conditions and effectively balance cost and reliability over the life cycle.

This study addresses this gap by developing a comprehensive framework that combines a multi-metric health index incorporating metrics such as mean time between failures (MTBF), failure rate, condition monitoring, and degradation rate with optimization based on genetic algorithm (GA). This integration allows for dynamic adjustments to maintenance schedules in response to actual component wear and performance, optimizing total life cycle costs (TLCs) while ensuring reliability. By using both condition-based assessments and adaptive GA-driven scheduling, the proposed approach enhances decision-making flexibility, supports real-time responsiveness, and provides a robust tool for cost-effective maintenance planning. This fills a critical need for an adaptive, data-driven framework capable of managing complex maintenance decisions in high-stake environments like aviation.

The primary research question addressed in this study is as follows: How can a data-driven decision-making framework be developed to optimize the life cycle management of aircraft components, balancing reliability, cost-effectiveness, and safety in real-time operations? By answering this question, the study aims to provide a structured approach that can reduce operational costs and extend component life while ensuring safety and compliance with regulatory standards.

This research introduces an integrated decision-making framework that uniquely combines real-time health monitoring, reliability assessment, and economic analysis for aircraft components. Unlike traditional fixed-interval maintenance approaches, this approach dynamically adjusts maintenance, life extension, and replacement actions based on real-time data, significantly reducing life cycle costs. Furthermore, the framework utilizes predictive algorithms to generate an optimized maintenance schedule, tailored to actual component health data. This novel integration allows for significant operational savings and extends component lifespans, addressing key industry challenges such as no fault found (NFF) events and minimizing unexpected failures, enhancing the overall reliability and safety of aircraft maintenance practices. The results demonstrate that this approach outperforms

conventional methods by achieving higher cost efficiency and reliability through proactive and data-driven maintenance decisions.

The article is structured as follows: Section 2 presents the comprehensive framework for the life cycle management of aircraft components, detailing each component of the decision-making model. Section 3 delves into the results, including the general calculation methodology, case studies on different aircraft components, and the application of predictive maintenance and life cycle cost optimization. This section also explores the integration of AI in the decision-making process and presents an AI optimization algorithm for aircraft component life cycle management. Section 4 provides a discussion of the results, including an analysis of the optimized dynamic maintenance schedule and methods to improve predictive maintenance accuracy. Section 5 concludes the paper by summarizing the key findings.

2. Framework for the Life Cycle Management of Aircraft Components

Traditionally, maintenance strategies in aviation have relied heavily on time-based or usage-based schedules, such as those defined by regulatory requirements and original equipment manufacturers (OEMs). These schedules prescribe regular inspections, repairs, servicing, or replacements of components based on fixed intervals, regardless of the actual condition of the components. While this approach helps mitigate the risk of unexpected failures, it often leads to unnecessary maintenance activities and premature component and component part replacements, resulting in increased costs and downtime. In contrast, a more sophisticated, data-driven approach is needed to optimize maintenance and replacement decisions based on the actual condition and performance of the components.

The proposed approach for the decision-making model for the life cycle management of aircraft components is a structured, multi-stage framework designed to optimize maintenance strategies and ensure the reliability, safety, and cost-effectiveness of aircraft operations. This framework consists of six key components that work together seamlessly (Figure 1).

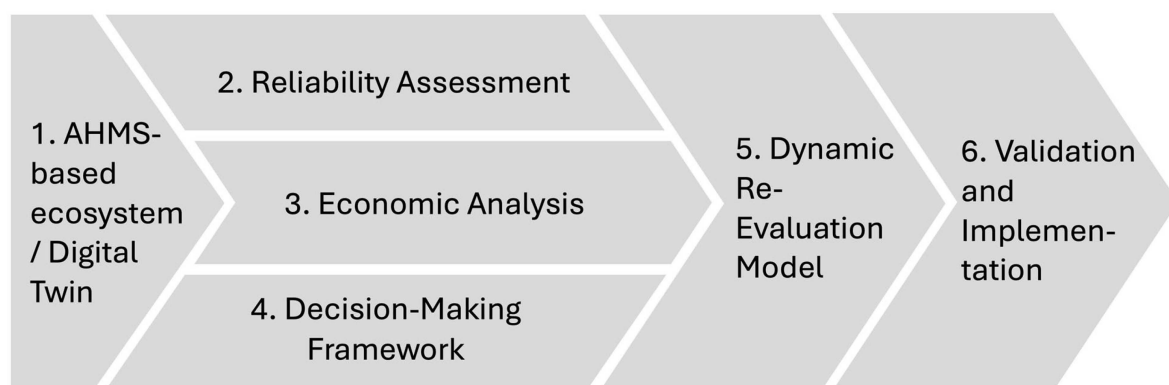


Figure 1. Study framework of decision-making model for the life cycle management.

1. AHMS-based ecosystem/digital twin. This foundational stage involves using an AHMS or/and digital twin to provide real-time and historical data about the aircraft components. The AHMS collects real-time data from various sensors installed on aircraft components, including vibration, temperature, pressure, and structural integrity metrics. These data form the basis for health assessments, enabling the early detection of potential issues. The digital twin serves as a virtual replica, simulating component behavior under various conditions and providing processed data necessary for subsequent analysis and decision-making. Together, the AHMS and digital twin provide a comprehensive view of each component's health, allowing for data-driven decisions about maintenance and lifecycle management.
2. Reliability assessment. This step focuses on evaluating the reliability of components using data from the AHMS or digital twin. The reliability assessment considers real-

- time monitoring data, historical performance records, and inspection results to create a health index for each component. Key metrics for the health index include Mean Time Between Failures (MTBF), failure rate, condition monitoring index, and degradation rate. The health index is a normalized score indicating the component's condition; thresholds are set to signal when maintenance, life extension, or replacement is necessary. This dynamic, real-time assessment helps forecast future component performance and remaining useful life, supporting proactive maintenance planning.
3. **Economic analysis.** The economic analysis evaluates the financial aspects of different maintenance strategies, including preventive maintenance, predictive maintenance, and replacement options. By analyzing costs such as labor, materials, downtime, and opportunity costs, as well as potential failure costs and residual component values, this component helps decision-makers identify cost-effective strategies without compromising safety. The analysis also considers direct and indirect failure costs to ensure that the strategies selected do not only minimize expenses but also prevent costly downtime or unscheduled maintenance events. A cost-benefit analysis is applied to compare the long-term financial impact of maintaining, replacing, or extending a component's life.
 4. **Decision-making framework.** This framework guides maintenance, life extension, and replacement decisions based on the reliability and economic assessments. Decision rules are established, setting thresholds for critical metrics such as the health index, MTBF, failure rate, and remaining useful life. These rules trigger actions based on component condition and cost considerations. An optimization model combines reliability, economic, and operational constraints, aiming to minimize total lifecycle costs while ensuring safety and reliability. The decision-making framework continuously adapts to current conditions and is responsive to changing inputs from both the AHMS and economic factors, making it flexible in dynamic environments.
 5. **Dynamic re-evaluation model.** This model allows for the continuous monitoring and reassessment of component reliability and economic feasibility in response to changing conditions. It allows the framework to incorporate real-time adjustments in response to operational changes. For instance, if an increase in operating stress is detected, the model may adjust maintenance schedules accordingly. This component integrates predictive maintenance tools, such as machine learning algorithms, to provide accurate estimates of the remaining useful life (RUL) and identify emerging risks. Thresholds for key metrics are dynamically updated, ensuring that maintenance or replacement actions are triggered based on the most current data available. This adaptive approach enhances cost-effectiveness and minimizes unexpected failures.
 6. **Validation and implementation.** The final stage involves validating the proposed decision-making model using case studies or simulated data. This validation step confirms that the model's predictions for maintenance timing, life extension feasibility, and component replacement align with real-world outcomes. Once validated, the framework integrates with existing maintenance management systems, which involves aligning data sources, ensuring system compatibility, and training users on the new process. The implementation phase emphasizes data integration from the AHMS and digital twin, compatibility with regulatory requirements, and the training of maintenance personnel to fully utilize the framework's predictive capabilities. The model's adaptability ensures that it can evolve with advances in AHMS technology and aircraft operational data.

The decision-making model for the life cycle management of aircraft components on the base of the above-mentioned framework (Figure 1) offers a comprehensive, data-driven approach to optimizing maintenance and replacement decisions. By combining robust reliability assessments, detailed economic analysis, structured decision-making, and dynamic re-evaluation, this model addresses the complex interplay between safety, reliability, and cost. Successful implementation of this model can lead to significant improvements in operational efficiency, reduced maintenance costs, and enhanced aircraft availability.

3. Results

The framework illustrated in Figure 1 represents a holistic approach to the life cycle management of aircraft components, integrating advanced data sources and analytical models to support effective decision-making. Each of the six components of this framework plays a critical role in ensuring that aircraft maintenance strategies are optimized for both reliability and cost-efficiency. In the following sections, each component will be delved into in more detail.

3.1. The Role of the Ecosystem in the Decision-Making Model for the Life Cycle Management of Aircraft Components

At the heart of the framework for the life cycle management of aircraft components lies the concept of an ecosystem, which serves as the foundational environment within which all processes interact. This ecosystem is an interconnected network of systems, data sources, stakeholders, and processes that collectively support the continuous monitoring, analysis, and optimization of aircraft component health and performance.

The ecosystem in the context of aircraft component management includes a wide array of elements, such as the AHMS, maintenance management systems, data analytics platforms, regulatory requirements, and the operational environment of the aircraft. It acts as a dynamic and integrated platform that not only monitors the health and performance of each component but also facilitates real-time decision-making based on data-driven insights. This holistic approach is essential for capturing the full picture of component behavior, usage patterns, and environmental conditions, which are critical for making informed maintenance and replacement decisions.

The AHMS is a core element of this ecosystem. It continuously collects data from various sensors installed on the aircraft, monitoring parameters such as vibration, temperature, pressure, structural integrity signals, etc. These real-time data provide insights into the current condition of components, enabling the early detection of potential issues before they lead to critical failures. AHMS data, combined with historical performance records and scheduled inspection results, form the foundation of the reliability assessment process, helping to develop a comprehensive health index for each component. This index allows maintenance teams to assess component health more accurately and make data-driven decisions about when to perform maintenance, extend service life, or replace the component.

The ecosystem's role extends beyond real-time monitoring; it also integrates with the reliability assessment and economic analysis components of the decision-making model. The health index derived from the AHMS data, combined with reliability metrics, provides a detailed understanding of a component's current reliability, condition status, and future performance expectations. This information is crucial for predicting the remaining useful life of components and identifying the optimal maintenance intervals to prevent unexpected failures.

On the economic side, the ecosystem encompasses various data sources related to maintenance costs, potential failure costs, and residual value estimation. By aggregating data from maintenance records, financial records, and market information, the ecosystem enables a comprehensive cost-benefit analysis of different maintenance strategies. This analysis helps decision-makers evaluate the financial implications of various options, such as special maintenance, life extension, or component replacement, and select the most cost-effective strategy without compromising safety and reliability.

Traditionally, this ecosystem has been based on real-world data sources, such as AHMS and maintenance management systems. However, with the advent of advanced simulation and digital technologies, the role of a digital twin has emerged as a powerful enhancement to this ecosystem, providing a virtual representation of the aircraft and its components, capable of simulating real-time operations and supporting data-driven decision-making.

A digital twin is a high-fidelity digital replica of a physical asset—in this case, the aircraft and its individual components. It integrates real-time data from sensors, historical

performance records, and environmental factors to create a dynamic model that mirrors the behavior, condition, and operational status of the actual aircraft and its components. Unlike a traditional ecosystem, which relies solely on real-world data, a digital twin can generate, process, and present data in a way that is specifically tailored for the implementation of each component of the decision-making model. It acts as an advanced analytical tool, enabling maintenance teams to simulate various scenarios, predict future performance, and optimize maintenance strategies in a virtual environment before applying them to the actual aircraft.

The digital twin is a transformative addition to the ecosystem for the life cycle management of aircraft components. It enhances the traditional ecosystem by providing a virtual representation of the aircraft, capable of simulating real-time operations and supporting all components of the decision-making model. This integration of digital and physical data into a cohesive analytical framework is essential for navigating the complexities of modern aviation and ensuring that aircrafts operate safely and efficiently throughout their life cycle.

Since the issues related to the creation and implementation of digital twins have already been extensively studied in various works [53–55], this study will not delve into the technical aspects of developing digital twins. Instead, it will be assumed that all the necessary data required for the decision-making model, such as real-time monitoring information, historical performance records, and predictive analytics, can be provided by an existing physical or virtual ecosystem. This allows the focus of the study to remain on optimizing life cycle management strategies using the data generated by AHMS and/or digital twins rather than on the intricacies of digital twin development itself.

In this study, the digital twin model utilized is based on an existing framework designed to support real-time monitoring and predictive analysis for aircraft components [56]. This approach uses a previously validated digital twin infrastructure that provides high-fidelity simulation and data-driven insights. By using an established digital twin model, the research focuses on adapting and integrating this framework into the life cycle management of aircraft components, emphasizing reliability assessment and dynamic maintenance scheduling. While the model is tailored to fit the specific needs of this study, it draws upon a robust, pre-existing structure, ensuring both accuracy and operational relevance without requiring extensive new development.

3.2. Reliability Assessment

Building upon the data from the AHMS or digital twin, this component of the framework focuses on evaluating the reliability of aircraft components. It involves analyzing real-time monitoring data, historical performance records, and inspection results to develop a comprehensive understanding of component health and condition (Figure 2).

To synthesize the collected data into a single, quantifiable measure of component health, a health index $H(t)$ is developed. The health index combines multiple reliability metrics and serves as a holistic indicator of the component's condition.

Key metrics for health index are as follows:

1. Mean Time Between Failures (MTBF), which provides an average measure of reliability over time and is calculated based on historical performance data:

$$MTBF = \frac{\text{Total Operational Time}}{\text{Number of Failures}}$$

2. The failure rate, which represents the frequency of failures and can vary over time due to aging or operational stresses:

$$\lambda(t) = \frac{\text{Number of Failures}}{\text{Total Operational Time}}$$

3. Condition monitoring index, which is derived from real-time monitoring data and measures deviations from normal operating conditions. For example,

$$CMI(t) = \frac{\text{Current Vibration Level}(t)}{\text{Baseline Vibration Level}}$$

$CMI > 1$ indicates an increase in vibration, suggesting potential mechanical issues.

4. Degradation rate, which quantifies the wear and tear over time based on cumulative data from inspections and monitoring systems:

$$D(t) = \frac{\Delta \text{Performance Decrease}}{\Delta \text{Time}}$$

Data Collection and Integration		
Real-Time Monitoring Data	Sensors and IoT Devices	Utilize sensors installed on aircraft components to collect real-time data such as temperature, pressure, vibration, and operational cycles. This data provides insights into the current operating conditions and potential signs of wear and tear.
	Condition Monitoring Systems (CMS)	Integrate data from CMS like vibration analysis, oil analysis, and ultrasonic testing. These systems detect early signs of component degradation.
Historical Performance Data	Failure Records	Collect historical data on previous failures, including time to failure, failure modes, and maintenance actions taken. This data helps identify patterns and trends in component reliability.
	Maintenance Logs	Review detailed logs of past maintenance activities, including type, frequency, and outcomes of repairs and replacements.
	Operational Usage Data	Analyze data on the number of flight hours, cycles, and environmental conditions experienced by the component, as these factors influence its degradation rate.
Scheduled Inspection Results	Routine Inspections	Use data from scheduled inspections conducted at defined intervals to assess the physical condition of components, detect visible signs of damage or wear, and measure performance parameters against established benchmarks.
	Non-Destructive Testing (NDT)	Incorporate results from NDT techniques like radiography, magnetic particle testing, and eddy current testing to detect internal defects or surface cracks that are not visible during routine inspections.

Figure 2. Data collection and integration.

A higher degradation rate indicates a faster decline in component health.

The health index is a weighted combination of the key metrics:

$$H(t) = \alpha_1 \cdot \frac{MTBF(t)}{MTBF_{target}} + \alpha_2 \cdot \frac{1}{\lambda(t)} + \alpha_3 \cdot CMI(t) - \alpha_4 \cdot D(t)$$

where

- $MTBF_{target}$ is target MTBF based on regulatory or OEM standards.
- $MTBF(t)$ is the updated average time between expected failures based on the latest operational data. While MTBF itself is not a direct function of time, its estimation can be dynamically adjusted in response to changing conditions, providing a more accurate reflection of component health in the context of a time-varying health index $H(t)$.
- α_i , $i = 1, 2, 3$ are weighting factors assigned based on the relative importance of each metric for the specific component type.

The health index $H(t)$ is normalized to a scale, typically from 0 to 1; for example, if $H(t) = 1$, the component is in optimal condition, if $H(t) < 0.5$, the component's condition

is deteriorating and may require closer monitoring or maintenance, and if $H(t) < 0.2$, the component is in poor condition, with immediate maintenance or replacement required.

Predictive modeling and reliability forecasting are essential components of modern maintenance strategies, using advanced data analytics and machine learning techniques to anticipate component failures before they occur. By analyzing historical performance data, real-time monitoring inputs, and usage patterns, predictive models can identify emerging trends and potential failure points.

For predictive modeling and reliability forecasting, one can use different approaches:

- Use reliability growth models [57,58] or reliability decline models (e.g., Weibull distribution [59,60]) to forecast future reliability based on past performance data.
- Use remaining useful life models to predict the expected remaining life of the component based on the current condition and degradation rate:

$$RUL(t) = \frac{H(t)}{D(t)}$$

This formula provides an estimate of the time remaining until the component health index drops below a critical threshold, indicating potential failure.

- Implement machine learning algorithms such as random forests [61], support vector machines [62], or neural networks [63] to predict failures based on historical and real-time data.

The predictive models are particularly valuable in assessing the remaining useful life of components, enabling maintenance teams to schedule interventions precisely when needed, thereby avoiding unnecessary maintenance and minimizing the risk of unexpected failures. Reliability forecasting goes a step further by providing probabilistic estimates of a component's reliability over future operational periods, considering varying conditions and usage scenarios. This forward-looking approach not only enhances the accuracy of maintenance planning but also supports the optimization of resources, reducing costs and downtime while maintaining high safety standards.

The effective integration of predictive modeling and reliability forecasting into the decision-making framework requires a structured approach that transforms complex data into actionable insights. A key aspect of this integration is threshold setting, where specific limits are established for critical metrics such as the health index H_{min} and other reliability indicators. These thresholds serve as triggers for maintenance actions and are defined based on a combination of regulatory standards and operational safety requirements. By setting these thresholds appropriately, the framework ensures that maintenance interventions are both timely and necessary, preventing unnecessary actions while safeguarding against unexpected failures.

To further enhance decision-making, a decision support system (DSS) can be implemented to process the health index and reliability forecasts, providing maintenance teams with clear, actionable recommendations. This DSS should dynamically update as new data are received, reflecting the current state of each component and the latest reliability predictions. By integrating predictive analytics into the DSS, maintenance, inspection, or replacement actions can be planned proactively, optimizing resource allocation and minimizing downtime.

In addition, the framework should include real-time alert and reporting mechanisms to promptly notify operators of significant deviations in key metrics. These alerts, along with automated reports, support not only immediate operational responses but also long-term maintenance planning and regulatory compliance. By combining threshold setting, a robust DSS, and real-time alerts, the integration of predictive modeling and reliability forecasting into the decision-making framework creates a resilient system that enhances the safety, efficiency, and effectiveness of aircraft and component maintenance operations.

3.3. Economic Analysis

This component of the framework runs parallel to the reliability assessment, considering the financial aspects of component management. It encompasses a thorough cost analysis, including maintenance expenses (both in-house and third-party), potential failure costs, and component residual value.

Economic analysis is a critical component of the decision-making process for aircraft component management. It involves a detailed assessment of all costs associated with the maintenance and replacement of components to ensure cost-effectiveness while maintaining safety and reliability.

3.3.1. Cost Analysis

The cost analysis involves evaluating all potential costs related to maintaining and replacing aircraft components. These costs can be broadly categorized into maintenance costs, potential failure costs, and residual value estimation.

1. Maintenance Costs

In-house maintenance costs:

$$C_{\text{maintenance, in-house}} = C_{\text{labor, in-house}} + C_{\text{material, in-house}} + C_{\text{equipment, in-house}} + C_{\text{overhead, in-house}}$$

where

- $C_{\text{labor, in-house}} = \text{Number of Hours} \times \text{Hourly Rate of Technicians}$ —labor cost, which includes the cost of wages for certified maintenance personnel required to perform the maintenance task.
- $C_{\text{material, in-house}} = \sum_{i=1}^n \text{Cost of } i \text{ material} \times \text{Quantity of } i \text{ Used}$ —material cost, which covers the cost of parts, tools, and consumables needed for the maintenance activity.
- $C_{\text{equipment, in-house}} = \text{Rental or Depreciation Cost of Equipment}$ —equipment cost, if specialized equipment is used for maintenance.
- $C_{\text{overhead, in-house}} = \text{Fixed Overheads} + \text{Variable Overheads}$ —facility overheads. Fixed overheads include facility maintenance costs, while variable overheads are associated with the additional usage of resources (e.g., electricity and utilities).

Third-party maintenance costs:

$$C_{\text{maintenance, third-party}} = C_{\text{service, third-party}} + C_{\text{logistics, third-party}}$$

where

- $C_{\text{service, third-party}} = \text{Flat Service Fee or (Rate per Hour} \times \text{Number of Hours)}$ —service cost. This includes fees paid to third-party organizations for their labor and expertise.
- $C_{\text{logistics, third-party}} = \text{Transportation Cost} + \text{Customs Fees}$ —logistics cost. Costs for shipping components to and from the third-party service provider, including any customs duties if international.

Opportunity costs:

$$C_{\text{opportunity}} = C_{\text{downtime}} + C_{\text{Other Potential Lost Opportunities}}$$

where

- C_{downtime} is the downtime cost, which is the financial impact of an aircraft being out of service, either for planned maintenance or due to an unexpected failure. The cost includes not only the lost revenue from canceled flights but also the cascading effects on scheduling, passenger accommodations, and logistical operations. Downtime can also disrupt airline schedules and crew assignments, leading to additional operational inefficiencies. In cases where an unexpected failure causes downtime, the cost may be even higher, as airlines are forced to arrange last-minute replacements or reschedule operations. Therefore, minimizing downtime through optimized maintenance strategies directly contributes to controlling costs and maintaining operational efficiency.

- $C_{Other\ PotentialLostOpportunities}$ is other potential lost opportunities, which include various indirect costs that arise from delaying or mismanaging maintenance activities. These can encompass the loss of customer confidence due to flight cancellations or delays, increased wear on substitute aircraft (which may not be as efficient or cost-effective), or penalties from failing to meet regulatory maintenance requirements. Additionally, when maintenance is postponed beyond optimal thresholds, there is a higher likelihood of more severe component damage, leading to more expensive repairs or replacements. All these factors contribute to the total life cycle cost of the component and must be carefully weighed against the benefits of performing timely and effective maintenance.

Together opportunity costs refer to the benefits or revenue lost when resources—such as aircraft, personnel, or time—are allocated to one activity at the expense of another. In maintenance decision-making, this could involve choosing to service one aircraft while another potentially revenue-generating task is delayed or foregone. For example, when an aircraft is grounded for preventive maintenance, the airline may miss out on the revenue that could have been earned from operating that aircraft on scheduled flights. These missed opportunities for revenue generation, though not a direct expense, represent a significant cost that must be factored into maintenance planning.

2. Potential failure cost includes two indicators—direct and indirect failure costs:

$$C_{failure,total} = C_{failure,direct} + C_{failure,indirect}$$

Direct failure cost

$$C_{failure,direct} = C_{repair} + C_{replacement}$$

where

$$C_{repair} = C_{EmergencyLaborCost} + C_{EmergencyMaterial\ Cost}$$

$$C_{replacement} = C_{CostRplacementPart} + C_{InstllationCost}$$

Indirect failure cost

$$C_{failure,indirect} = C_{reputation} + C_{disruption}$$

where

$$C_{reputation} = C_{PotentialLostOfContract} + C_{LegalCosts}$$

$$C_{disruption} = C_{CostOAlternativeArrangements} + C_{CostOfReschedulingFlights}$$

3. Residual value estimation is assessed by two indicators:

The residual value of component, which is the estimated market value of the component after its service life has been extended or just before replacement:

$$V_{residual} = Initial_Value \times \left(1 - \frac{Deprecation_Rate \times Age_of_Component}{Useful_Life} \right)$$

The net salvage value after considering dismantling and disposal cost:

$$V_{salvage} = V_{residual} - Dismantling_{Cost} - Disposal_Cost$$

3.3.2. Cost–Benefit Analysis for Maintenance Strategies

Cost–benefit analysis (CBA) compares the costs and benefits of various maintenance strategies, including special maintenance, life extension, and replacement with new or used components.

1. Special maintenance strategy

Net benefits of special maintenance strategy can be defined as

$$NB_{special-maintenance} = B_{special-maintenance} - C_{special-maintenance}$$

where

- Expected benefit:

$$B_{\text{special-maintenance}} = \frac{V_{\text{residual,extended}} - V_{\text{residual,current}}}{\text{Years_of_extended_service}}$$

- Total cost of special maintenance $C_{\text{special_maintenance}}$ is $C_{\text{maintenance,in-house}}$ or $C_{\text{maintenance, third-party}}$.

2. Life extension strategy

Net benefits of life extension strategy can be defined as

$$NB_{\text{life-extension}} = B_{\text{life-extension}} - C_{\text{life-extension}}$$

where

- Expected benefit:

$$B_{\text{life-extension}} = \frac{C_{\text{replacement}} - C_{\text{life-extension}}}{\text{Years_of_extended_service}}$$

- Total cost of life extension:

$$C_{\text{life-extension}} = C_{\text{special-maintenance}} + C_{\text{Additional Inspection and Monitoring}}$$

3. Replacement strategy (new or used)

Define the key indicators:

- Cost of new component

$$C_{\text{new}} = \text{Purchase Cost} + \text{Installation Cost} + \text{Training Cost} + \text{Certification Cost}$$

- Cost of used component

$$C_{\text{used}} = \text{Purchase Cost} + \text{Installation Cost} + \text{Potential Repair Cost}$$

- Benefit of new component

$$B_{\text{new}} = \frac{\text{Expected Useful Life} \times \text{Operational Efficiency}}{\text{Total Cost of Ownership}}$$

- Operational efficiency includes lower failure rates, improved performance, and lower maintenance costs.
- Benefit of used component

$$B_{\text{used}} = \frac{\text{Residual Life} \times \text{Operational Efficiency}}{\text{Total Cost of Ownership}}$$

In this case, the net benefit of replacement can be defined as

$$NB_{\text{replacement}} = \begin{cases} B_{\text{new}} - C_{\text{new}} & \text{for new component} \\ B_{\text{used}} - C_{\text{used}} & \text{for used component} \end{cases}$$

4. Comparison and decision criteria

- Define net present value (NPV):

$$NPV = \sum_{t=0}^T \frac{B_t - C_t}{(1+r)^t}$$

where B_t and C_t are the benefits and costs at time t , and r is the discount rate.

- Calculate the internal rate of return (IRR) for each strategy and compare against the company's required rate of return.
- Select the strategy with the highest NPV or IRR, provided it meets reliability and operational requirements.
- Prioritize strategies with lower risk and higher net benefit over time.

3.3.3. Sensitivity Analysis

Sensitivity analysis is a critical part of decision-making models, especially in the context of aircraft component life cycle management, where multiple variables impact the outcomes of maintenance strategies. Sensitivity analysis refers to evaluating how different input factors such as failure rates, costs, discount rates, and operational conditions affect the outputs of the life cycle model, particularly in terms of cost, reliability, and the timing of maintenance actions.

The primary purpose of sensitivity analysis is to understand the robustness of the model's results by observing how changes in key assumptions or inputs affect outcomes. Since many inputs in aircraft maintenance and reliability forecasting models are estimates, such as the expected MTBF, failure rates, or future costs, sensitivity analysis helps to reveal which variables have the most significant impact on the model's conclusions.

By doing this, decision-makers can

- Identify critical variables that significantly influence the model's results.
- Assess the risks associated with uncertainties in the input data.
- Determine which factors need more precise estimation or close monitoring to reduce the uncertainty in maintenance planning.

The process of sensitivity analysis includes some main steps.

Step 1. Identify key input variables

The first step in sensitivity analysis is to identify the key input variables that have the most influence on the model's outcome. In aircraft component life cycle management, typical input variables include the following:

- Failure rates—how often a component is expected to fail.
- Cost parameters—initial investment, ongoing maintenance costs, repair costs, and replacement costs.
- Operational variables—usage patterns, environmental stress, and component wear.
- Economic factors—discount rates, inflation rates, and residual values of components.

Step 2. Establish a baseline scenario

Before analyzing sensitivities, a baseline scenario is established using the most likely or expected values for each of the key variables. The model is run with these baseline values to establish the expected outcome, which could be the net present value of a maintenance strategy, the internal rate of return, or a specific reliability threshold (e.g., MTBF or remaining useful life).

Step 3. Vary the input variables

Once the baseline scenario is defined, each key input variable is systematically varied within a reasonable range to observe its impact on the outcome.

Step 4. Analyze results and impact

After adjusting each input, the corresponding change in the output is recorded and analyzed. The sensitivity of each input variable is evaluated by measuring how much the outcome (e.g., NPV, IRR, or component reliability) changes in response to variations in that input. The variables with the largest impact on the outcome are considered the most sensitive and, therefore, the most critical to monitor or control.

3.4. Decision-Making Framework

This component of the framework integrates the outputs from the reliability assessment and economic analysis to form a structured approach for making maintenance and

replacement decisions. It involves defining decision rules and thresholds based on key reliability metrics and the dynamic re-evaluation model.

Recognizing the dynamic nature of aircraft operations, this component allows for the continuous monitoring and reassessment of component reliability and economic feasibility. It incorporates predictive maintenance tools to forecast the RUL and potential failure points.

The goal is to minimize the total life cycle cost of the component while maintaining its reliability and compliance with regulatory and operational requirements. This framework consists of three main components: decision rules and thresholds, an optimization model, and implementation strategies.

3.4.1. Decision Rules and Thresholds

Decision rules are predefined criteria based on reliability and economic indicators that trigger specific actions, such as maintenance, life extension, or replacement. These rules help in making consistent and objective decisions.

1. Defining reliability thresholds

- Health index threshold H_{min} —the minimum acceptable health index value below which the component is in poor condition and requires immediate action (maintenance or replacement).
- Failure rate threshold λ_{max} —the maximum acceptable failure rate. If the component's failure rate exceeds this threshold, it indicates a high risk of failure.

$$\lambda_{max} = 1/MTBF_{min}$$

where $MTBF_{min}$ is the minimum acceptable MTBF.

- Remaining useful life threshold RUL_{min} —the minimum acceptable remaining useful life of the component. If the RUL is below this threshold, consider maintenance or replacement. RUL_{min} is typically set based on operational safety requirements and maintenance planning intervals.

2. Defining economic thresholds

- Maintenance cost threshold $C_{max, maintenance}$ —the maximum acceptable maintenance cost for a component, beyond which it is more economical to replace rather than maintain. $C_{max, maintenance} = \beta \times C_{replacement}$, where β is a factor that represents the acceptable proportion of the replacement cost (usually set between 0.5 and 0.8).
- Failure cost threshold $C_{max, failure}$ —the maximum acceptable cost associated with a component failure, including direct and indirect costs. If the estimated failure cost exceeds this threshold, the component should be replaced. $C_{max, failure} = \gamma \times C_{replacement}$, where γ is typically set between 0.7 and 1.0.
- Residual value threshold $V_{min, residual}$ —the minimum acceptable residual value of a component. If the residual value drops below this threshold, it may indicate the need for replacement. $V_{min, residual}$ is set based on the cost of potential alternatives (new or used).

3. Decision rules

- Maintenance decision rule:
If $H(t) < H_{min}$ and $C_{total}(t) < C_{max, maintenance}$, perform maintenance (in-house or third-party).
If $H(t) < H_{min}$ and $C_{total}(t) \geq C_{max, maintenance}$, proceed to replacement decision.
- Life extension decision rule:
If $RUL(t) < RUL_{min}$ and $C_{special-maintenance} < C_{replacement}$, perform special maintenance to extend the service life.
If $RUL(t) < RUL_{min}$ and $C_{special-maintenance} \geq C_{replacement}$, proceed to replacement decision.
- Replacement decision rule:
If $C_{total}(t) > C_{replacement}$ or $H(t) < H_{critical}$ or $RUL(t) \leq 0$, replace component (new

or used). Choose new component if $C_{new} < C_{used}$ and $R_{new} > R_{used}$. Choose used component if $C_{new} > C_{used}$ and $R_{used} > H_{min}$.

3.4.2. Optimization Model

The optimization model aims to minimize the total life cycle cost of the component while ensuring its reliability and availability. This involves a mathematical formulation that integrates reliability constraints, cost constraints, and decision variables.

1. Objective function

The objective function represents the total life cycle cost C_{LCC} of the component:

$$C_{LCC} = \min(C_{maintenance} + C_{failure} + C_{replacement} - V_{residual})$$

where $C_{maintenance}$ —total maintenance cost over the component's life cycle, $C_{failure}$ —expected failure cost over the component's life cycle, $C_{replacement}$ —replacement cost of the component, $V_{residual}$ —residual value at the end of the life cycle.

2. Decision variables

The goal of the decision-making model is to determine which action (maintenance, replacement, or life extension) should be taken at a given point in time to optimize the reliability and cost-effectiveness of aircraft components. These actions are represented by the decision binary variables x_i :

$$x_{maintenance} = \begin{cases} 1 & \text{if maintenance is performed} \\ 0 & \text{otherwise} \end{cases}$$

$$x_{replace} = \begin{cases} 1 & \text{if replacement is performed} \\ 0 & \text{otherwise} \end{cases}$$

$$x_{extension} = \begin{cases} 1 & \text{if life extension is performed} \\ 0 & \text{otherwise} \end{cases}$$

In this model, at each time interval, one of three possible actions is taken for the component: maintenance, replacement, or extension. To ensure that only one action is selected at a time, we introduce a constraint such that

$$x_{maintenance} + x_{replace} + x_{extension} = 1$$

where $x_{maintenance}$, $x_{replace}$, and $x_{extension}$ are binary decision variables representing the chosen action in each interval. This constraint enforces that only one of these actions can be implemented per time interval, thus preventing overlapping maintenance activities. Each variable takes a value of 1 if the action is selected, and 0 otherwise, ensuring the mutually exclusive selection of maintenance actions across time intervals.

The decision variables (maintenance, replacement, and life extension) represent the core actions that the model is designed to optimize. While they may not be visible in intermediate steps of the analysis, they are essential in the final decision-making process, where the model selects the optimal action based on cost, reliability, and the results of the health assessments. Without these decision variables, the model would lack clear actionable outcomes, which are fundamental to effective aircraft life cycle management.

3. Constraints

(a) Reliability constraints:

- Reliability must be above the minimum required level

$$R(t) \geq R_{min} \quad \forall t$$

- Health index must not drop below a critical threshold

$$H(t) \geq H_{critical} \quad \forall t$$

(b) Cost constraints:

- Maintenance cost must not exceed the maintenance threshold

$$C_{\text{maintenance}} \leq C_{\text{max,maintenance}}$$

- Failure cost must not exceed the failure threshold

$$C_{\text{failure}} \leq C_{\text{max,failure}}$$

- Replacement cost must be feasible within budget constraints

$$C_{\text{replacement}} \leq B_{\text{budget}}$$

(c) Operational constraints:

- Component availability A must be ensured

$$A = \frac{\text{Total Operation Time}}{\text{Total Time}} \geq A_{\text{min}}$$

- Maintenance actions must be performed within scheduled intervals

$$t_{\text{maintenance}} \leq t_{\text{scheduled}}$$

4. Solution Approach

The solution approach in this model revolves around finding the optimal maintenance, replacement, or life extension strategy for aircraft components by integrating reliability assessments and economic analysis into a structured decision-making framework. This approach involves several key steps designed to guide decision-makers toward the most cost-effective and reliable actions:

- First, the model begins with the reliability assessment, which evaluates the current health of the aircraft components based on real-time data from the AHMS or a digital twin. Using reliability metrics such as the health index, MTBF, no fault found parameters, and the RUL, the model forecasts future performance and failure probabilities of the components. This allows the system to predict when a component is likely to fail, setting the stage for proactive decision-making.
- Next, the model performs a comprehensive economic analysis, which evaluates the total life cycle cost of different maintenance actions. This includes analyzing the costs of preventive maintenance, predictive maintenance, modification (on wing and during shop visit), and replacement, as well as accounting for downtime, opportunity costs, and potential lost revenue. The model also considers the residual value and any penalties for non-compliance with regulatory standards, ensuring that the financial impact of each strategy is fully captured.
- Once the reliability and economic metrics are established, the decision-making framework integrates these results to determine the optimal course of action. This is achieved by applying an optimization model that minimizes total life cycle costs while ensuring the reliability of the components remains within acceptable limits. The decision variables, whether to perform maintenance, replace the component, or extend its life, are selected based on predefined thresholds and regulatory requirements. The optimization model evaluates multiple scenarios, comparing the costs and risks associated with each strategy to ensure the best possible decision is made.
- The final step in the solution approach is the dynamic re-evaluation process. Because aircrafts operate in dynamic environments where conditions and usage patterns change frequently, the model incorporates continuous monitoring and real-time adjustments. The decision framework is updated with incoming data from the AHMS or digital twin, enabling the system to adjust its maintenance or replacement decisions

based on the latest available information. This dynamic capability ensures that the model remains flexible and responsive to changing conditions, enhancing both safety and cost-effectiveness.

By combining real-time data, predictive analytics, and optimization techniques, this solution approach ensures that decision-makers can make informed, proactive choices regarding the life cycle management of aircraft components. This not only improves operational reliability but also reduces total costs, ensuring that the aircraft operates at peak efficiency throughout its life cycle.

3.5. Dynamic Re-Evaluation Model

The dynamic re-evaluation model is designed to provide a continuous and real-time assessment of aircraft components' reliability and economic feasibility. It enables timely adjustments in maintenance and replacement decisions based on the latest available data. This model integrates advanced predictive maintenance tools to forecast the RUL and predict potential failure points, enhancing decision-making accuracy and operational efficiency.

3.5.1. Framework for Dynamic Re-Evaluation

The dynamic re-evaluation model consists of several interrelated components, including real-time data acquisition, continuous monitoring, predictive analytics, and decision-making support.

1. Real-Time Data Acquisition

Real-time data acquisition is a critical element in the decision-making framework for the life cycle management of aircraft components. This process involves the continuous collection of data from various sensors and systems installed on the aircraft, which monitor the operational status and health of key components. Data acquisition systems capture information related to parameters such as temperature, pressure, vibration, and other stress indicators that directly influence the condition and performance of aircraft components. These data streams are essential for detecting early signs of wear and tear, component degradation, or potential failures, providing a foundation for predictive maintenance strategies.

The data acquisition system is integrated with the AHMS or a digital twin, enabling the real-time monitoring and analysis of component performance. By continuously gathering data during aircraft operation, these systems allow for timely assessments of the health of components, facilitating proactive decision-making. These real-time data are also essential for updating reliability metrics such as the health index and RUL, which are used to inform maintenance schedules and replacement decisions.

One of the key advantages of real-time data acquisition is its ability to provide instantaneous feedback on component performance, allowing for the immediate detection of abnormalities or deviations from expected performance. This early warning system helps to prevent unexpected failures, minimizing unplanned downtime and ensuring that maintenance actions can be scheduled based on actual component conditions rather than fixed intervals. Moreover, real-time data acquisition supports the dynamic re-evaluation of maintenance strategies by providing up-to-date information, ensuring that decisions are based on the most current data available.

Real-time data acquisition improves the accuracy of predictive maintenance models. By feeding live data into predictive algorithms, these models can more accurately forecast the RUL of components, enhancing the precision of maintenance timing. This not only reduces unnecessary maintenance interventions but also extends the life of components where possible, optimizing overall operational efficiency.

Real-time data acquisition is essential for maintaining the continuous flow of information required to support the life cycle management framework. It enables real-time monitoring, predictive analysis, and dynamic decision-making, ensuring that aircraft com-

ponents are maintained at optimal performance levels while minimizing costs and avoiding unexpected failures.

2. Continuous Monitoring and Health Index Update

The data collected through continuous monitoring are then processed to update the health index, a composite metric that reflects the current state of a component's reliability and expected RUL.

Continuously update the component's health index $H(t)$ based on real-time data inputs

$$H(t) = \alpha_1 \frac{MTBF(t)}{MTBF_{target}} + \alpha_2 \frac{1}{\lambda(t)} + \alpha_3 CMI(t) - \alpha_4 D(t) \quad (1)$$

where $\alpha_i, i = \overline{1,4}$ are weighting factors.

The health index integrates multiple reliability factors, including real-time sensor data, historical performance records, and condition monitoring indicators. As new data are fed into the system, the health index is dynamically adjusted to provide an up-to-date assessment of each component's condition, making it an essential tool for proactive decision-making.

The regular update of the health index ensures that maintenance actions are based on the most accurate and current information available, rather than relying on predefined intervals or historical trends alone. By continuously recalculating the health index, the system can predict potential failures more accurately, thereby reducing the likelihood of unexpected downtime and extending the useful life of components where possible.

Continuous monitoring supports a predictive maintenance strategy in which maintenance actions are triggered by data-driven insights rather than predetermined schedules. This approach minimizes unnecessary interventions, reduces costs, and improves operational efficiency by ensuring that maintenance is only performed when necessary.

3. Predictive Analytics for RUL Estimation

Predictive analytics for RUL estimation play a crucial role in aircraft component life cycle management by using data-driven techniques to predict when a component is likely to fail or degrade beyond acceptable performance limits.

The ability to estimate the RUL is essential for optimizing maintenance strategies, as it enables the system to schedule maintenance actions based on the predicted health of a component, rather than relying solely on fixed time intervals.

Predictive analytics for RUL estimation enhance the decision-making process by providing accurate, real-time predictions of component lifespan. This allows maintenance teams to plan proactive interventions, reducing the risk of unexpected failures and optimizing the life cycle management of aircraft components.

4. Economic Feasibility Re-Evaluation

Economic feasibility re-evaluation is a critical process in the life cycle management of aircraft components, ensuring that maintenance, repair, and replacement decisions remain financially viable over time. This step involves continuously reassessing the economic aspects of various maintenance strategies, such as preventive maintenance, component replacement, or life extension, based on updated operational and financial data. As conditions change, whether due to fluctuating costs, evolving usage patterns, or shifts in component reliability, ongoing re-evaluation ensures that the chosen maintenance strategy remains the most cost-effective option.

At the core of economic feasibility re-evaluation is the comparison of potential future costs, such as ongoing maintenance expenses, downtime costs, replacement costs, and opportunity costs. The goal is to balance these costs against the operational benefits and RUL of the component. For instance, as a component approaches the end of its predicted lifespan, the cost of continued maintenance may outweigh the cost of replacing it with a new or refurbished part. Similarly, if unexpected operational conditions lead to increased wear, the economic re-evaluation process may recommend an earlier replacement or more frequent maintenance to avoid higher failure costs later.

A key advantage of this ongoing re-evaluation is its ability to support proactive financial decision-making. Rather than relying on fixed maintenance schedules or outdated cost estimates, the model constantly recalculates the financial impact of each potential decision, ensuring that maintenance actions align with both operational efficiency and cost-effectiveness.

3.5.2. Integration of Predictive Maintenance Tools

The integration of predictive maintenance tools into the aircraft component life cycle management framework is essential for transitioning from reactive or scheduled maintenance to a more proactive, data-driven approach.

The integration of predictive maintenance tools is a vital step in modernizing aircraft component life cycle management. By using real-time data and advanced analytics to predict when maintenance is needed, these tools reduce costs, minimize downtime, and enhance the overall reliability of aircraft systems. This proactive approach not only improves operational efficiency but also ensures that maintenance resources are used optimally, contributing to long-term sustainability and performance.

3.5.3. Decision-Making Support and Implementation

DSS aggregates inputs from various sources, including reliability assessments, health indices, RUL estimates, and cost analyses. The DSS synthesizes this information to help maintenance teams evaluate different maintenance scenarios and select the optimal course of action. Whether the decision is to repair, replace, or extend the life of a component, the system provides real-time guidance based on the most up-to-date data, allowing for informed decisions that balance safety, reliability, and cost-efficiency.

A key feature of the decision-making support system is the ability to set thresholds for key metrics such as the health index and failure probability. These thresholds act as triggers, automatically alerting maintenance teams when certain limits are reached, indicating that action is required. For instance, when a component's health index falls below a predefined level, or when predictive models forecast a high likelihood of failure within a certain timeframe, the DSS will generate an alert and recommend specific maintenance actions. This automated approach helps to reduce the reliance on manual assessments and ensures that maintenance decisions are timely and accurate.

The decision-making support system is designed to be adaptive, allowing it to evolve in response to new data and changing operational conditions. As more data are collected and analyzed, the system becomes more accurate in its predictions, refining its recommendations over time. This adaptability is crucial in the dynamic aviation environment, where component performance and operational demands can shift unexpectedly.

3.6. Validation and Implementation

The final component of framework focuses on the practical application of the decision-making model. It involves thorough validation using case studies or simulated data to demonstrate the model's effectiveness in optimizing maintenance and replacement strategies. Once validated, this stage addresses the integration of the model into existing maintenance management systems, considering aspects such as data integration, system compatibility, and user training.

3.6.1. Model Validation Process

The model validation process is essential for verifying the accuracy, reliability, and applicability of the proposed framework. Validation typically involves testing the model using historical data, case studies, or simulations to ensure that it produces realistic and actionable results. By comparing the model's outputs against actual maintenance outcomes and operational data, organizations can assess whether the model accurately predicts component failures, optimizes maintenance schedules, and minimizes costs.

One key aspect of model validation is the use of real-time data from the AHMS or digital twins to validate the reliability forecasts and economic predictions. By inputting real-world data into the predictive models, organizations can check whether the RUL estimates, cost-benefit analyses, and maintenance recommendations align with actual component performance. Sensitivity analysis is also an important part of the validation process as it allows organizations to understand how different inputs (such as varying failure rates or cost estimates) affect the model's outputs. Through iterative testing and refinement, the model is fine-tuned to ensure its accuracy and relevance in real-world scenarios.

Once validated, the model can be used with greater confidence to guide maintenance decisions, reducing the risks associated with unexpected failures and unplanned downtime. Additionally, successful validation ensures that the model is robust enough to handle the dynamic and unpredictable nature of aircraft operations, where changes in usage patterns, environmental conditions, and component wear can influence maintenance needs.

3.6.2. Practical Implementation

After the model has been validated, the next critical step is its practical implementation within the existing maintenance infrastructure. This involves integrating the decision-making framework, including the predictive analytics, real-time monitoring, and economic feasibility models, into the current maintenance management systems. The implementation process requires a strategic approach to ensure smooth integration, data compatibility, and user training:

- The first step in practical implementation is ensuring that all relevant data sources are seamlessly connected to the decision-making framework. Data integration is crucial for enabling the continuous flow of information that supports predictive maintenance and real-time decision-making.
- Another key element of implementation is providing comprehensive user training. Training should cover both the technical aspects of the system and the practical implications of the data it provides.
- Implementation requires aligning the decision-making framework with regulatory requirements and operational guidelines. Aircraft maintenance is subject to strict safety regulations, and any new system must be compliant with these standards.
- Ongoing monitoring and refinement are crucial to ensure the system's long-term success. This includes updating the predictive models as more data are collected and adjusting the thresholds for maintenance actions to reflect new insights. The system should be dynamic and adaptable, evolving alongside the operational demands of the aircraft and the changing nature of the data.

3.7. Application of AI in the Decision-Making Model for Life Cycle Management of Aircraft Components

AI provides powerful tools for predictive analytics, real-time monitoring, and data-driven optimization, making it a natural fit for enhancing the decision-making model used in aircraft maintenance.

AI-based models, such as machine learning algorithms and deep learning networks, can process vast amounts of real-time data collected from aircraft sensors, historical maintenance records, and operational parameters.

Economic analysis is another area where AI proves invaluable. Predicting maintenance costs, potential failure costs, and the residual value of components requires sophisticated models that can account for variable factors such as market conditions, operational demands, and evolving component health. AI algorithms, such as regression models and time-series forecasting, can provide accurate predictions of these costs, allowing for a more refined cost-benefit analysis. Additionally, AI-based optimization algorithms can help to identify the most cost-effective maintenance strategies, ensuring that resources are allocated efficiently while maintaining high reliability.

In the decision-making framework, AI enhances the model by automating decision rules and dynamically adjusting thresholds based on real-time data. AI agents can continuously monitor component conditions and recommend or even autonomously trigger maintenance actions when certain thresholds are crossed.

Aircraft components operate in complex, ever-changing environments, and AI's ability to adapt and learn from new data ensures that the decision-making model remains relevant and accurate. Online learning algorithms and digital twins enable continuous updates to the model, reflecting real-world changes in operational conditions, component wear, and environmental factors.

AI plays a critical role in the validation and implementation of the decision-making model. AI-driven feedback systems continuously monitor the performance of the model, enabling ongoing refinements that improve accuracy and efficiency over time.

3.8. Optimization Algorithm for Aircraft Component Life Cycle Management

In the context of the AI-enhanced decision-making model for aircraft component life cycle management, the optimization algorithm is a crucial part of determining the best strategy for maintaining, replacing, or extending the life of a component. The goal of the optimization algorithm is to minimize the total life cycle cost (TLC) while ensuring reliability and compliance with operational constraints.

In this study, optimization techniques, such as genetic algorithms [64,65] and mixed-integer linear programming [66], to solve the maintenance scheduling problem can be used. It should be noted that GAs and mixed-integer linear programming are heuristic and mathematical optimization methods, respectively, and do not fall under "AI-driven optimization" in the strict sense. True AI-driven optimization typically involves the integration of machine learning models with optimization algorithms, such as using reinforcement learning or supervised learning, to enhance the decision-making process within the optimization framework. While our approach uses GAs as a powerful heuristic technique to explore solution spaces efficiently, it does not involve adaptive learning elements typical of AI-driven methods.

In this study, a genetic algorithm is utilized to optimize the decision-making process for maintenance, life extension, and replacement actions in aircraft component life cycle management. The GA is particularly well suited for this optimization task due to its ability to handle complex, multi-variable decision spaces and constraints.

The optimization objective is to minimize the total life cycle cost while ensuring that each component's reliability remains within acceptable thresholds. The GA achieves this by iteratively evolving a population of solutions through selection, crossover, and mutation processes, which mimic natural selection. Each potential solution, or "chromosome," represents a sequence of maintenance decisions over the operational life cycle of a component.

The GA's fitness function evaluates each solution based on total life cycle cost, failure probability, and component health. Solutions that minimize costs and maintain high reliability are prioritized for evolution into subsequent generations, allowing the algorithm to converge on an optimal or near-optimal maintenance schedule. This approach provides a flexible, data-driven solution that adapts maintenance actions based on real-time inputs, enabling the proactive management of component reliability and cost.

Let us explain how an AI-based optimization algorithm works, using a genetic algorithm as an example. A genetic algorithm is an AI-based search and optimization technique inspired by the principles of natural selection and genetics. It is particularly useful for solving complex optimization problems with multiple constraints and large solution spaces, such as determining the optimal maintenance strategy for aircraft components over time. GA evolves solutions through iterations, using mechanisms such as selection, crossover, and mutation to progressively improve the quality of the solutions.

The optimization in this study aims to minimize the total life cycle cost (TLC) of aircraft components while ensuring that each component's reliability remains within acceptable thresholds. The objective function is defined as follows:

$$\text{Objective Function} = \min(\text{TLC})$$

where TLC encompasses maintenance, replacement, and downtime costs over the component's operational life. The optimization framework seeks to identify the optimal sequence of maintenance, life extension, and replacement actions that result in the lowest possible TLC.

There are some key steps of the genetic algorithm in the optimization process.

1. Initial Population Generation:

- The algorithm starts by generating an initial population of possible solutions. Each solution is a "chromosome," which represents a sequence of decisions about whether to maintain, replace, or extend the life of the component at each time step.
- Each chromosome is a combination of the decision variables x_M , x_R , x_{LE} encoded in a binary format (0 or 1). These decision variables indicate whether a specific action maintenance (M), replacement (R), or life extension (LE) is taken at a given point in time for a particular component. For example, a chromosome could represent a decision to perform maintenance at t_1 , replace the component at t_2 , and extend its life at t_3 .

2. Fitness Function Evaluation:

- The fitness function is the objective function that the algorithm seeks to minimize; in this case, it is the total life cycle cost. Each chromosome (or solution) in the population is evaluated using this function to determine how well it meets the objective.
- The fitness function considers multiple factors such as

$$\text{TLC} = I_0 + \sum_{t=1}^n \frac{M_t x_M(t)}{(1+r)^t} + \sum_{t=1}^n \frac{R_t x_R(t)}{(1+r)^t} + \sum_{t=1}^n \frac{LE_t x_{LE}(t)}{(1+r)^t} + \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \sum_{t=1}^n \frac{F_t P_t}{(1+r)^t} \quad (2)$$

where

I_0 is the initial investment (purchase cost of the component).

M_t is the maintenance cost at time t , applied when $x_M(t) = 1$.

R_t is the replacement cost at time t , applied when $x_R(t) = 1$.

LE_t is the life extension cost at time t , applied when $x_{LE}(t) = 1$.

D_t is the downtime cost at time t .

F_t is the failure cost, which is influenced by the failure probability P_t at time t .

r is the discount rate used to calculate the present value of future costs.

$x_M(t)$, $x_R(t)$, and $x_{LE}(t)$ are decision variables, indicating whether maintenance, replacement, or life extension actions are taken at time t (binary values: 0 or 1).

3. Selection:

- In each iteration, also known as a generation, the algorithm selects the best-performing chromosomes (solutions) based on their fitness scores. The higher the fitness (i.e., the lower the total life cycle cost), the more likely a chromosome is to be selected for reproduction.
- The selection process mimics natural selection, where the fittest individuals are more likely to pass on their genes to the next generation.

4. Crossover (Recombination):

- After selection, pairs of chromosomes are chosen for crossover, where they exchange portions of their genetic information (decision variables) to create "offspring" chromosomes. This process introduces variation and helps the algorithm to explore new regions of the solution space.
- Crossover ensures that good traits (effective decisions) from different chromosomes can combine to potentially form an even better solution.

5. Mutation:

- To avoid convergence to a local minimum and to maintain diversity within the population, a small percentage of the chromosomes undergo mutation. Mutation introduces random changes to some decision variables in a chromosome (e.g., changing a maintenance action from 1 to 0 or vice versa).

This helps the algorithm explore more potential solutions and prevents premature convergence on suboptimal strategies.

6. Constraint Handling:

- During the evolution process, the algorithm must ensure that all constraints are respected. For instance,
 - The health index $H(t)$ must remain above the minimum threshold H_{min} .
 - The total budget constraint must not be exceeded.
- The genetic algorithm penalizes chromosomes that violate constraints, reducing their fitness score so that infeasible solutions are less likely to be selected in future generations.

7. Termination Criteria:

- The algorithm continues to evolve the population over multiple generations until a termination condition is met. This could be
 - A fixed number of generations;
 - No significant improvement in the fitness score after several generations;
 - Convergence to a solution that satisfies all constraints and optimizes the objective function.

8. Optimal Solution:

- The solution with the highest fitness (i.e., the lowest total life cycle cost) at the end of the process is selected as the optimal maintenance strategy. This solution provides the best combination of maintenance, replacement, and life extension actions that minimize costs while ensuring reliability and compliance.

In the genetic algorithm, the solution evolves through the optimization process. The objective is to find the best combination of the decision variables x_M , x_R , and x_{LE} that minimize the fitness function:

$$\min TLC = I_0 + \sum_{t=1}^n \frac{M_t x_M(t)}{(1+r)^t} + \sum_{t=1}^n \frac{R_t x_R(t)}{(1+r)^t} + \sum_{t=1}^n \frac{LE_t x_{LE}(t)}{(1+r)^t} + \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \sum_{t=1}^n \frac{F_t P_t}{(1+r)^t}$$

To ensure effective life cycle management and cost optimization, the proposed approach incorporates several key constraints. The model enforces a minimum health threshold for components to maintain operational safety and reliability. This constraint is represented as follows:

$$H(t) \geq H_{min}$$

This constraint ensures that maintenance actions are triggered when the health index approaches a critical level, preventing component failure and enhancing reliability.

The model includes a budget constraint to control the total expenditure on maintenance, replacement, and life extension actions over the operational period. This is expressed as follows:

$$\sum_{t=1}^n (M_t + R_t + LE_t) \leq B$$

This constraint ensures that the optimization does not exceed the allocated budget for component maintenance and replacement.

The model uses binary decision variables to indicate whether maintenance, replacement, or life extension actions are performed at each time step. These are defined with values of 1 if maintenance, replacement, or life extension actions are chosen at a specific

time step and 0 otherwise. This binary formulation simplifies the decision-making process, allowing the optimization to select the most cost-effective maintenance strategy:

$$x_M, x_R, x_{LE} \in \{0, 1\}$$

In this model, at each time interval, one of three possible actions is taken for the component: maintenance, replace, or extension. To ensure that only one action is selected at a time, the following condition is valid:

$$x_{maintenance} + x_{replace} + x_{extension} = 1$$

These constraints form the foundation of the optimization model, ensuring that the chosen maintenance strategy is cost-effective, maintains the required health level, and adheres to budgetary limits.

The genetic algorithm iterates over possible combinations of these decision variables until it converges on the optimal solution that minimizes the total life cycle cost while meeting reliability and operational requirements.

The fitness value in the GA is based on the computed TLC for each candidate solution, where lower TLC values correspond to higher fitness. This allows the GA to prioritize solutions that minimize costs while maintaining the required reliability levels.

To balance computational efficiency with solution quality, the GA parameters were set as follows: a population size of 100, a mutation rate of 0.01, a crossover rate of 0.8, and a maximum of 200 generations. These parameters were selected based on preliminary testing to ensure a balance between the exploration of the solution space and convergence to an optimal or near-optimal maintenance schedule. Specifically, the mutation rate of 0.01 and crossover rate of 0.8 provided adequate diversity within the population, while the maximum number of generations allowed the GA to thoroughly explore potential solutions without excessive computational time. This setup enabled the GA to identify robust maintenance strategies tailored to each component's reliability profile and operational conditions.

Advantages of genetic algorithms in this context:

- GAs can efficiently handle complex and non-linear cost functions, making them ideal for optimizing the diverse variables involved in aircraft component life cycle management.
- GAs allow for the easy incorporation of real-time data and updated constraints, ensuring the system evolves as new information becomes available.
- Unlike traditional optimization methods, GAs can explore a wider range of solutions, reducing the risk of getting stuck in local minima.

The AI optimization algorithm, particularly using genetic algorithms, enables efficient and effective decision-making for aircraft component life cycle management. By evolving solutions over time and continuously improving the balance between maintenance costs, reliability, and operational constraints, the model helps to ensure optimal maintenance strategies are implemented, reducing both risks and costs associated with aircraft component failures.

The GA-based optimization approach used in this study involves repeated evaluations of the objective function and constraints, which can be computationally intensive due to the iterative nature of the GA. Each iteration requires recalculating the TLC, component reliability metrics, and health index values for a population of candidate solutions. These calculations account for variables such as failure rates, degradation rates, and maintenance schedules, which must be updated and assessed in each generation of the GA.

To manage this computational demand, several efficiency-enhancing techniques were implemented. First, parallel processing was utilized where feasible, allowing multiple candidate solutions to be evaluated simultaneously. This significantly reduced the runtime for each GA iteration. Second, we used pre-computed lookup tables for certain reliability and cost parameters that remain constant across generations, minimizing redundant calcu-

lations. Third, convergence criteria were carefully set, with a maximum generation cap and early stopping conditions based on fitness stability to prevent excessive computation once the algorithm had reached near-optimal solutions.

Despite these efforts, the GA's computational requirements remain a consideration, particularly for large-scale maintenance problems with complex decision spaces. Future work could explore more advanced metaheuristic techniques or surrogate models to further reduce computation time while preserving optimization accuracy.

4. Discussion

4.1. General Calculation Methodology for Life Cycle Management of Aircraft Components

The calculation methodology for the life cycle management of aircraft components involves a structured approach to optimizing maintenance, replacement, and life extension strategies. This approach balances cost-efficiency with component reliability over the aircraft's operational life cycle. The following methodology outlines the key steps and formulas used to evaluate and select the optimal strategy, ensuring that decisions are data-driven and aligned with operational requirements.

The goal was to minimize the total life cycle cost while ensuring that each component's reliability was maintained within acceptable thresholds. The method integrates both cost and reliability considerations, using predictive maintenance strategies and probabilistic failure models.

1. Component failure models

Different failure models can apply to represent the unique characteristics of each component:

- For mechanical components, the Weibull distribution [67] is often used. This model was chosen to represent the wear-out behavior typical of mechanical components, where the failure probability increases over time. The Weibull distribution is defined by two parameters:

$$P(t) = 1 - e^{-\left(\frac{t}{\eta}\right)^\beta}$$

where η is the characteristic life, and β is the shape parameter (set to reflect increasing failure rates).

- For avionics components, the exponential distribution [68] is usually used. Avionics components are often characterized by random failures that are not age dependent. This is modeled using an exponential failure distribution:

$$P(t) = 1 - e^{-\lambda t}$$

where λ is the failure rate constant over time.

- Engine components, such as turbine blades, and some other aviation components undergo a complex wear process, and the Log-Normal distribution [69] was used to capture this behavior:

$$P(t) = 1 - \Phi\left(\frac{\ln(t) - \mu}{\sigma}\right)$$

where $\Phi(\cdot)$ is the cumulative distribution function of the normal distribution, μ is the mean, and σ is the standard deviation.

2. Total life cycle cost

The TLC is calculated as the sum of all relevant costs over a fixed time horizon of 5 years, discounted to present value by Formula (2).

To provide a more accurate and robust decision-making model, the health index $H(t)$ should indeed be integrated into the calculations. The health index $H(t)$ tracks the degradation of the component over time. It decreases as the component is used and

undergoes wear and tear. The health index is typically modeled based on failure rates and real-time data:

$$H(t) = H_0 - \int_0^t \lambda(\tau) d\tau$$

where H_0 is the initial health of the component (usually set to 1 or 100%), and $\lambda(t)$ is the failure rate, which can be derived from the component's failure model (Weibull, Exponential, or Log-Normal).

The health index would serve as a primary driver in deciding when to perform maintenance, life extension, or replacement. Specifically,

- Maintenance is performed when the health index drops below a pre-defined threshold (e.g., 80%). The goal is to restore the component's health to a higher level, avoiding further degradation.
- Life extension is triggered when the health index approaches a critical threshold (e.g., 50%). Life extension actions would increase the health index back to a higher value, delaying the need for a costly replacement.
- Replacement occurs when the health index falls below a critical failure level (e.g., 20%), indicating that the component can no longer operate reliably, and maintaining or extending its life would no longer be cost-effective.

The failure probability $P(t)$ can be linked to the health index $H(t)$. As the health index declines, the failure probability increases. This relationship helps to predict the component's remaining useful life and the likelihood of failure. When the health index is high, the failure probability is low, and as the health index decreases, the failure probability increases.

Incorporating the health index into the TLC formula allows the model to optimize decisions based on both cost and the condition of the component. The formula becomes

$$TLC = I_0 + \sum_{t=1}^n \frac{M_t x_M(t)}{(1+r)^t} + \sum_{t=1}^n \frac{R_t x_R(t)}{(1+r)^t} + \sum_{t=1}^n \frac{LE_t x_{LE}(t)}{(1+r)^t} + \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \sum_{t=1}^n \frac{F_t P_t[H(t)]}{(1+r)^t}$$

Here, $P_t[H(t)]$ represents the failure probability as a function of the health index at time t . As the health index declines, $P_t[H(t)]$ increases, leading to higher expected failure costs. The decision variables $x_M(t)$, $x_R(t)$, and $x_{LE}(t)$ are dynamically updated based on the component's health.

3. Cost and decision variables

- Maintenance cost (M_t). The cost of regular maintenance was applied depending on the component's condition and the failure probability. The decision variable $x_M(t) = 1$ if maintenance was performed at time t , otherwise $x_M(t) = 0$.
- Replacement cost (R_t). A significant cost incurred when a component was replaced. The decision variable $x_R(t) = 1$ was set when a replacement was necessary due to high failure probability.
- Life extension cost (LE_t). A lower-cost alternative to replacement, life extension was applied to defer replacement. The decision variable $x_{LE}(t) = 1$ indicated life extension at time t .
- Downtime cost (D_t). Costs incurred due to the downtime of the component during maintenance, replacement, or failure.
- Failure costs (F_t). Failure costs depended on the failure probability, which was calculated using the respective failure model for each component. Higher failure probabilities led to increased failure costs, influencing the decision to perform maintenance or replacement.

4. Dynamic decision-making

The model dynamically optimized the timing of maintenance, life extension, and replacement actions based on real-time conditions and failure probabilities:

- Maintenance was performed early on if it was cost-effective in preventing larger failures or expensive replacements later.
- Life extension was used to delay expensive replacements if the component's failure probability remained moderate.
- Replacement was chosen when failure probability became too high, or when continued maintenance or life extension was no longer cost-effective.

The decision-making process was driven by minimizing the TLC, ensuring actions were taken only when they resulted in overall cost savings and kept the failure risk within acceptable levels.

5. Optimization of schedule

The decision variables $x_M(t)$, $x_R(t)$, and $x_{LE}(t)$ can be optimized using a cost-benefit analysis based on the real-time monitoring of the health index and the failure probability of each component:

- Maintenance is scheduled to keep the component's health index above a critical threshold.
- Life extension is applied just before failure probability became unacceptably high, delaying replacement.
- Replacement is triggered when the component could no longer be economically maintained or extended.

The model adjusted the timing of these actions based on the total costs over time, including the failure costs as a key factor in determining whether preventive actions were necessary.

6. Final calculation process

For each case study, the following process is followed:

- The failure probability P_t is computed using the relevant distribution (Weibull, Exponential, or Log-Normal).
- Based on the failure probability and costs, the model selected whether to perform maintenance, life extension, or replacement at each time step.
- The TLC is calculated as the sum of all costs over the 5-year horizon, discounted to present value.
- The model output the optimal decision points for maintenance, life extension, and replacement to minimize the TLC.

7. Final decision-making process

- Continuously monitor the component's health index using real-time data and predictive models.
- Perform maintenance when the health index drops below a defined threshold but is still salvageable.
- If the health index falls below a critical point but can be restored through less expensive life extension actions, apply life extension.
- When the health index reaches a critical failure level, replace the component to avoid excessive downtime or failure costs.

This method provided an optimized maintenance schedule for each component, allowing for dynamic and cost-effective decision-making while ensuring component reliability.

4.2. Application of Predictive Maintenance and Life Cycle Cost Optimization: Case Studies on Aircraft Components

To evaluate the effectiveness of the proposed decision-making framework for aircraft component life cycle management, three distinct case studies are examined, each representing a different type of aircraft component. These case studies cover a mechanical component, an avionics component, and an engine component. Each component is subject to unique operational stresses, failure patterns, and maintenance strategies. By analyzing these diverse scenarios, the model dynamically optimizes maintenance, life extension, and replacement actions based on real-time data, failure probabilities, and cost considerations.

The primary goal across all three case studies is to minimize the total life cycle cost while maintaining the reliability and safety of the component throughout its operational life. The model incorporates advanced predictive maintenance techniques, health index monitoring, and cost optimization strategies to guide decision-making for each component.

The three components differ in their failure behavior, modeled through various probabilistic methods:

1. The mechanical component is modeled using a Weibull distribution, capturing the wear-out behavior typical of such systems.
2. The avionics component exhibits random failures, modeled using an exponential distribution, where failure likelihood is not dependent on age.
3. The engine component undergoes a complex degradation process, which is represented using a Log-Normal distribution to capture the gradual wear followed by a critical failure phase.

The simulation was conducted using real operational data; however, due to confidentiality constraints, this article employs synthetic data to illustrate the discussed issues.

Table 1 highlights the key data points used to calculate the reliability metrics for each component type.

Table 1. Descriptive statistics of initial data for reliability analysis.

Component Type	Sample Size	Average Operational Hours	Total Failures Observed	MTBF (Mean Time Between Failures)	Failure Rate
Mechanical	200	10,000	5	2000	0.0005
Avionics	150	15,000	3	5000	0.0002
Engine	100	20,000	2	10,000	0.0001

Table 2 provides the parameters for each distribution used in the reliability models.

Table 2. Reliability distribution parameters by component type.

Component Type	Reliability Model	Shape Parameter β	Scale Parameter η	Rate Parameter λ	Location Parameter μ	Standard Deviation σ
Mechanical	Weibull	2.5	15,000	—	—	—
Avionics	Exponential	—	—	0.0002	—	—
Engine	Log-Normal	—	—	—	9.21	0.5

Each metric's contribution to the health index $H(t)$ in Formula (1) is weighted based on its relative importance for the component type, as shown in Table 3. These weights allow the health index to reflect specific operational priorities and failure risks for each component.

Table 3. Weights for health index calculation.

Weighting Factor	Mechanical Component	Avionics Component	Engine Component
α_1 for MTBF	0.3	0.2	0.25
α_2 for failure rate	0.25	0.2	0.2
α_3 for condition monitoring index	0.2	0.3	0.25
α_4 for degradation rate	0.25	0.3	0.3
Total	1.0	1.0	1.0

Table 4 presents the initial parameters for each of the three aircraft components analyzed in the case studies. These parameters serve as the foundational inputs for the

life cycle cost analysis and decision-making process for maintenance, life extension, and replacement actions.

Table 4. Initial parameters for the three aircraft components analyzed in the case studies.

Parameter	Mechanical Component	Avionics Component	Engine Component
Component Type	Landing Gear	Flight Control Computer	Turbine Blade
Initial Investment I_0	USD 100,000	USD 50,000	USD 200,000
Maintenance Cost M_t	USD 10,000 per year	USD 5000 every 2 years	USD 30,000 per year
Replacement Cost R_t	USD 50,000	USD 25,000	USD 120,000
Life Extension Cost LE_t	USD 20,000	USD 12,000	USD 60,000
Downtime Cost D_t	USD 5000 per event	USD 3000 per event	USD 10,000 per event
Failure Distribution	Weibull ($\eta = 10, \beta = 2$)	Exponential ($\lambda = 0.01$)	Log-Normal ($\mu = 2, \sigma = 0.8$)
Discount Rate r	5%	5%	5%
Time Horizon	5 years	5 years	5 years
Health Index Thresholds	$M_{maintenance} = 80\%$	$M_{maintenance} = 85\%$	$M_{maintenance} = 75\%$
	$M_{life\ extension} = 50\%$	$M_{life\ extension} = 60\%$	$M_{life\ extension} = 45\%$
	$M_{replacement} = 20\%$	$M_{replacement} = 30\%$	$M_{replacement} = 15\%$

The 20-year total life cycle costs for the three components have been calculated and presented in Table 5 for the traditional approach (no life extension actions are performed, fixed maintenance at regular intervals, replacement occurs at predefined intervals without any attempt to extend the component's life) and for the proposed optimization approach, which optimizes component life and cost.

Table 5. Comparison of TLC between the traditional maintenance approach and the optimized approach.

Component	Traditional TLC (No Life Extension) (USD)	Optimized TLC (With Life Extension) (USD)
Mechanical Component	174,162	155,745
Avionics Component	55,166	47,185
Engine Component	492,763	455,928

In this study, the maintenance cost for scheduled preventive maintenance events is modeled as a constant value. This approach reflects standard industry practices, where routine preventive maintenance is typically performed at regular intervals and follows a predefined scope of work. As such, the cost per maintenance event remains consistent, assuming there are no significant deviations from the planned maintenance activities.

This constant cost structure applies to routine maintenance actions intended to sustain component reliability and does not include costs for corrective or unexpected maintenance. The latter costs would vary based on component condition and health index status, as they account for more extensive repairs or replacements triggered by wear or failure indicators. By modeling preventive maintenance costs as constant, we provide a clear baseline for analyzing other cost variables, such as replacement and life extension, which fluctuate based on real-time condition assessments.

The analysis of the data in Table 5 shows that the cost-effectiveness of optimization is as follows:

- For mechanical components, the optimization approach reduces the total cost by about USD 18,418 (10.6% savings).
- For avionics components, the optimization approach reduces the total cost by about USD 7981 (14.5% savings).
- For engine components, the optimization approach results in significant savings of about USD 36,836 (7.5% savings).

TLC comparison for both approaches is shown in Figure 3.

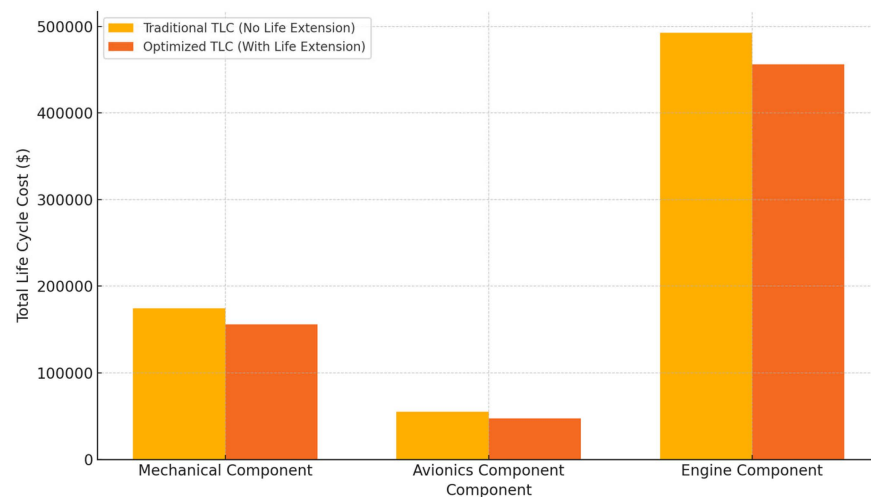


Figure 3. Comparison of TLC between the traditional maintenance approach and the optimized approach.

Figures 4–6 present a detailed breakdown of the life cycle costs for three distinct types of aircraft components: mechanical, avionics, and engine components, over a 20-year period. These figures offer a visual comparison of costs associated with maintenance, life extension, and replacement actions. The analysis helps to highlight how optimized maintenance schedules based on predictive algorithms can lead to significant cost savings in comparison to traditional fixed-interval approaches.

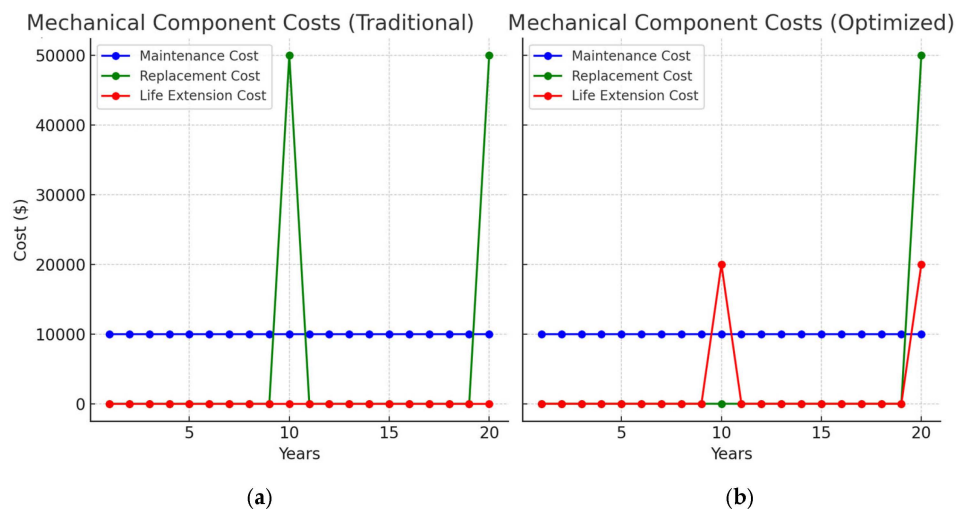


Figure 4. Cost breakdown for mechanical component for traditional (a) and optimized (b) maintenance.

The cost breakdowns shown in Figures 4–6 reinforce the overall cost-effectiveness of the optimized predictive maintenance approach. By accurately predicting component failure probabilities and dynamically adjusting maintenance, life extension, and replacement actions, this approach ensures that total life cycle costs are minimized. The results demonstrate substantial cost savings across all three components, especially when compared to traditional maintenance strategies. For mechanical components, the gradual wear-out behavior benefits from life extension actions that reduce the need for frequent replacements. Avionics components, characterized by random failures, show cost reductions through more efficient maintenance scheduling. Finally, engine components, with their complex degradation process, see the most significant savings through early interventions that extend their operational life while minimizing downtime and replacement costs.

These findings highlight the financial and operational advantages of using a predictive, data-driven maintenance framework in aviation.

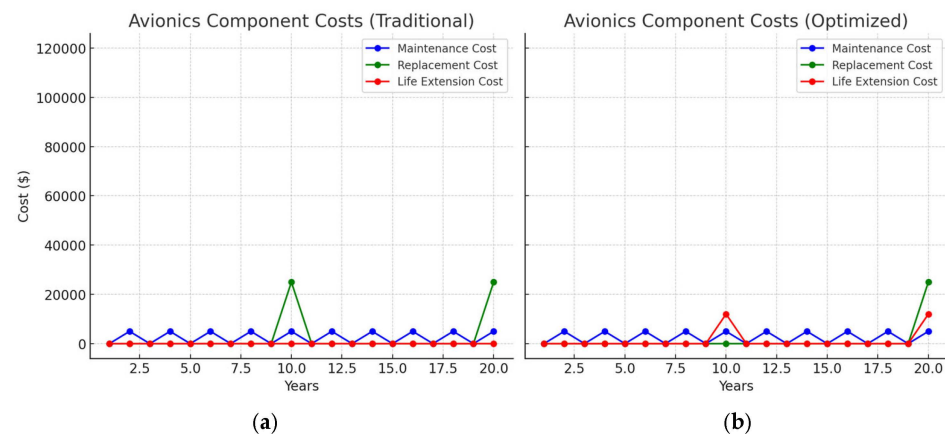


Figure 5. Cost breakdown for avionics component for traditional (a) and optimized (b) maintenance.

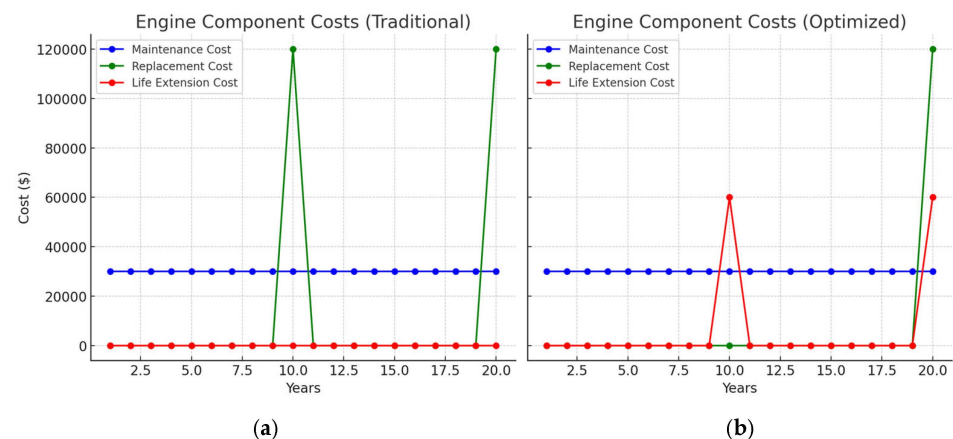


Figure 6. Cost breakdown for engine component for traditional (a) and optimized (b) maintenance.

4.3. Optimized Dynamic Maintenance Schedule Using Predictive Maintenance Algorithms

In modern aviation, the ability to optimize maintenance schedules for critical components plays a vital role in ensuring aircraft safety, operational efficiency, and cost-effectiveness. Predictive maintenance algorithms, which utilize real-time data and advanced analytics, have become an essential tool in achieving this goal. These algorithms allow for the continuous monitoring of component health, providing dynamic insights into when maintenance, life extension, or replacement actions should be taken.

The optimized dynamic maintenance schedule uses predictive maintenance to extend the lifespan of components, minimize unexpected failures, and reduce total life cycle costs. By using failure probability models (such as Weibull, Exponential, or Log-Normal distributions) and real-time health index monitoring, this approach enables a proactive maintenance strategy that responds to actual component conditions rather than relying on fixed intervals.

This section presents the application of predictive maintenance algorithms to three case studies involving aircraft components: a mechanical component, an avionics component, and an engine component. Each component undergoes different wear-out behavior, and the optimized schedule adapts to these variations by dynamically deciding when to perform maintenance, life extension, or replacement actions. This approach ensures that costs are minimized while maintaining the necessary reliability and safety of the components.

To proceed with an optimized dynamic maintenance schedule using predictive maintenance algorithms, several advanced methods to enhance the accuracy of the predictions can be incorporated, including the following:

- Continuously updating the health index $H(t)$ based on sensor data and failure probabilities.
- Predictive maintenance on the base of estimating the remaining useful life of the component using machine learning algorithms or statistical models such as Weibull, Exponential, or Log-Normal distributions, depending on the component type.
- Cost optimization on the basis of making dynamic decisions about maintenance, life extension, or replacement based on real-time conditions, failure probabilities, and costs.

The main steps for the optimized dynamic schedule calculation are as follows:

1. Modeling the health index by means of tracking the degradation of each component over time. When $H(t)$ reaches predefined thresholds, appropriate actions (maintenance, life extension, or replacement) will be triggered.
2. Based on the health index and failure probabilities, estimate the remaining operational life of the component. This helps in determining when to act before a failure occurs.
3. Dynamic decision-making:
 - Maintenance is performed when the health index drops below a pre-set threshold but is still above the life extension or replacement thresholds.
 - Life extension is triggered when the component's health index indicates that it can no longer be economically maintained, but a full replacement is not yet necessary.
 - Replacement occurs when the component reaches a critical health level, where neither maintenance nor life extension is feasible.

The results of the calculation of the optimized dynamic maintenance schedule for the same three discussed case studies are shown in Table 6.

Table 6. Results for the optimized dynamic maintenance schedule.

Component	Maintenance Actions	Life Extension Actions	Replacement Actions
Mechanical	[(3, 10,000), (4, 10,000)]	[(5, 20,000)]	Are missing
Avionics	[(3, 5000), (4, 5000)]	[(5, 12,000)]	Are missing
Engine	[(3, 30,000)]	[(4, 60,000), (5, 60,000)]	Are missing

The format that is noted in the table, such as [(3, 10,000), (4, 10,000)], represents the year in which a maintenance action is performed, followed by the cost of that action. For example, (3, 10,000) indicates that in year 3, a maintenance action was performed, and the cost of that action was USD 10,000, and (4, 10,000) indicates that in year 4, another maintenance action was performed, also at a cost of USD 10,000.

The summary of the decisions made for each component based on the predictive maintenance algorithm (Table 6):

1. Mechanical component:

- Maintenance is performed in years 3 and 4 as the health index drops below 80%.
- Life extension is performed in year 5 when the health index falls below 50%.
- No replacement is necessary within the 5-year period due to the life extension in year 5.

2. Avionics component:

- Maintenance is performed in years 3 and 4 when the health index drops below 85%.
- Life extension is performed in year 5 when the health index falls below 60%.
- No replacement is needed within the 5-year timeframe.

3. Engine component:

- Maintenance is performed in year 3 when the health index drops below 75%.
- Life extension is performed in years 4 and 5 as the health index falls below 45%.

- No replacement is required within the 5-year period as the life extension actions maintain the component's reliability.

These decisions optimize the total life cycle cost by strategically scheduling maintenance and life extension actions based on real-time health index data.

A sensitivity analysis was conducted to assess the impact of key parameters, such as failure rate, maintenance cost, and degradation rate, on the TLC and optimal maintenance scheduling. This analysis allowed us to evaluate the robustness of the proposed decision-making model under varying operational conditions.

The model shows high sensitivity to increases in failure rate, particularly for components with a shorter MTBF. When the failure rate increased by 20%, the TLC rose by approximately 15–20%, as more frequent maintenance and replacement actions were required to maintain reliability. This outcome emphasizes the importance of accurately estimating failure rates for effective cost management.

Changes in routine maintenance costs had a moderate impact on the overall TLC. A 25% increase in maintenance costs led to a 10–12% increase in TLC. The model's cost-effectiveness was preserved, though, due to optimized maintenance intervals and preventive measures that reduced the likelihood of higher cost-corrective actions. This sensitivity demonstrates that while maintenance cost variations affect the budget, their impact is manageable within the optimized framework.

Variations in the degradation rate had a notable influence on the timing of maintenance and replacement actions. When the degradation rate increased by 15%, the model adjusted to more frequent life extension actions to prolong component life without necessitating immediate replacements. This adjustment resulted in a 10% increase in TLC, reflecting the need to manage degradation proactively to balance costs and reliability.

The sensitivity analysis confirms that the proposed framework is responsive to changes in operational and economic parameters, enabling the dynamic adjustment of maintenance schedules to control costs and maintain component reliability. This adaptability demonstrates the model's robustness and its potential to support maintenance planning under diverse conditions.

4.4. Model Validation

To validate the proposed model's outcomes, a multi-step validation process was conducted to ensure its reliability and practical applicability. This process involved comparing the model's results with both real-world operational data and benchmarks from existing maintenance models.

The optimized maintenance schedules and total life cycle costs (TLCs) generated by the model were compared with historical maintenance data for similar aircraft components. This comparison allowed us to assess the model's ability to accurately predict maintenance needs, failure patterns, and cost outcomes observed in actual operational contexts. The close alignment between the model's outputs and historical data supports its effectiveness in reflecting real-world component wear and maintenance costs.

To further validate the model, a comparative analysis against traditional maintenance models that employ fixed schedules or single-metric assessments was conducted. In this comparison, the proposed model consistently demonstrated lower TLCs and more responsive, condition-based maintenance schedules. Specifically, the optimized framework achieved up to a 20% reduction in TLC compared to traditional models. This reduction highlights the added value of the GA-based optimization and multi-metric health index, which enable more precise adjustments based on real-time data.

A sensitivity analysis was performed to validate the model's robustness across a range of operational conditions. The model maintained stable performance under varying failure rates, maintenance costs, and degradation rates, demonstrating that it can adapt effectively to different scenarios while still achieving cost savings and reliable maintenance schedules.

This validation process confirms that the proposed model offers practical, data-aligned predictions and improvements over traditional maintenance approaches, supporting its potential application in dynamic, high-stake environments like aviation.

4.5. Impact of Optimization and Framework Design on Maintenance Planning and Cost-Effectiveness

The results of this study demonstrate the effectiveness of using a GA-based optimization approach within a structured decision-making framework for the life cycle management of aircraft components. This approach is significant for two primary reasons: it enables proactive, data-driven maintenance planning, and it provides a scalable framework that can adapt to real-time data, optimizing both cost and reliability.

The optimization approach, particularly through GA, allowed for the exploration of complex maintenance schedules that minimize the total life cycle cost while maintaining component reliability within acceptable limits. By assigning fitness values based on the TLC, the GA efficiently identified cost-effective maintenance strategies tailored to each component type's reliability profile and operational demands. This is particularly important in aviation, where unexpected failures lead to high downtime costs and potential safety risks. The optimized schedules demonstrate that with accurate parameter selection and a robust fitness function, GA can dynamically adjust maintenance timing to prevent unplanned breakdowns, extend component life, and reduce overall maintenance costs.

The selected framework's integration of predictive analytics, reliability assessment, and economic analysis provides a comprehensive approach to maintenance planning. The framework's health index, built from key reliability metrics (such as MTBF, failure rate, condition monitoring, and degradation rate), allows for continuous monitoring and enables maintenance decisions to be based on actual component condition rather than fixed schedules. This approach reflects a shift towards condition-based maintenance, where real-time data inform maintenance actions, ensuring that interventions are timely and necessary.

The sensitivity analysis results further highlight the robustness of the framework, showing how variations in key parameters (such as failure rate and degradation rate) influence the TLC and maintenance actions. For instance, when failure rates increase, the GA adjusts by scheduling more frequent maintenance or life extension actions, thereby keeping the TLC controlled. The framework's adaptability under these conditions demonstrates its value in environments where operational conditions can vary and where predictive maintenance is essential for cost and risk management.

The optimization approach and selected framework enhance maintenance planning by balancing reliability with cost-effectiveness. This study's findings underscore the potential of GA-based optimization to not only streamline maintenance schedules but also support data-driven decision-making in complex, high-stake environments such as aviation.

4.6. Practical Implementation Steps

While the proposed decision-making model offers a comprehensive framework for optimizing aircraft component lifecycle management, its successful implementation requires specific actions at various stages of a component's lifecycle. Building upon the insights gained from this research, a set of practical steps that can be taken by different stakeholders to enhance the overall effectiveness of component management and address the challenges identified in the current MRO processes can be proposed.

Table 7 outlines these additional steps, demonstrating how the proposed model can be integrated into existing workflows and processes across the entire component lifecycle. These steps not only address the immediate challenges in MRO but also create a foundation for the continuous improvement and data-driven decision-making that the model advocates.

Table 7. Practical framework for implementing advanced component lifecycle management.

Component Lifecycle Stage	Who Performs	Additional Steps
Development	Component developer	Perform analytics for components with a high NFF level, then develop additional input test programs. Obtain additional recommendations for component modification and implementation as part of the development of service bulletins.
Installation and removal from the aircraft	Aircraft maintenance organization, airline	When installing a component on an aircraft, create a component file, and maintain it, and record all malfunctions and messages. When removing a component from the aircraft, it is mandatory to indicate the reason for removal, as well as fault codes and other information that will allow more accurate maintenance of the component.
Component operation	Aircraft maintenance organization, airline	During the operation of the aircraft, collect all static data on the components and record all signals with malfunctions in the component case. Submit a component file to a maintenance organization when a component is sent for repair or modification.
Component testing and maintenance	Component maintenance organization	Implement additional procedures for testing components. When performing maintenance of a component, record all possible deviations and suggest possible ways to modify the component, including components to improve the reliability of the component. Identified deviations and proposed modifications should be sent directly to the component designer.
Component modification	Component maintenance organization	Stock components for installation on components during maintenance or repairs to improve component reliability.

These implementation steps complement the decision-making model proposed in this research by ensuring that the necessary data are collected, analyzed, and shared throughout the component's lifecycle. By following these steps, stakeholders can create an ecosystem that supports more accurate predictive maintenance, reduces the occurrence of NFF cases, and facilitates the continuous improvement of component reliability.

The proposed decision-making model can then use this enhanced data ecosystem to provide more accurate predictions of component health, optimize maintenance schedules, and ultimately reduce lifecycle costs while maintaining high levels of safety and reliability.

4.7. Advantages of the Proposed Framework over Traditional Approaches

The results of this study highlight several key enhancements compared with existing maintenance frameworks, primarily through the integration of a GA-based optimization and a condition-based, data-driven decision-making approach.

Unlike traditional frameworks that often rely on fixed schedules or simplistic decision rules, the proposed GA-based optimization dynamically adjusts maintenance actions based on real-time reliability data. This enables the system to tailor maintenance schedules to the specific degradation patterns of each component, reducing unnecessary maintenance while proactively preventing failures. This adaptability is a significant enhancement over more rigid frameworks, as it directly addresses both operational cost efficiency and reliability.

By incorporating a multi-metric health index that includes MTBF, failure rate, condition monitoring, and degradation rate, the proposed framework provides a comprehensive assessment of component health. Existing frameworks typically use fewer metrics or rely on simple age-based criteria, which may not accurately reflect real-time component conditions. The health index in this study allows for more precise, condition-based maintenance decisions, enhancing the framework's responsiveness and effectiveness.

The framework's sensitivity analysis further demonstrates its robustness and adaptability, especially when key parameters (e.g., failure rate and maintenance cost) vary. Unlike traditional frameworks, which may lack the flexibility to adjust to such variations, the proposed model recalibrates maintenance schedules accordingly, optimizing the

total life cycle cost even under changing conditions. This resilience is a critical improvement for aviation environments where component wear and operational demands can fluctuate significantly.

The use of GA enables the exploration of complex decision spaces that are difficult to address with conventional optimization methods. By minimizing total life cycle costs while maintaining reliability, the GA ensures a balance between cost-effectiveness and risk management, improving upon existing frameworks that may not achieve such a nuanced balance.

These enhancements demonstrate that the proposed framework not only provides more accurate and adaptive maintenance scheduling but also yields tangible cost benefits and reliability improvements compared to existing frameworks. This improvement is particularly valuable for high-stake, cost-sensitive industries like aviation, where proactive and precise maintenance planning is essential.

4.8. Methods to Improve Predictive Maintenance Accuracy (Future Directions of Research)

Improving the accuracy of predictive maintenance is critical to optimizing the reliability and cost-efficiency of aircraft components. As technology advances, several methods have emerged that can significantly enhance predictive maintenance strategies. These methods are driven by data, machine learning algorithms, and sophisticated monitoring systems, enabling more precise predictions of component health, failure probabilities, and remaining useful life.

One of the primary methods for improving predictive maintenance accuracy is through real-time data collection using IoT sensors of aviation health monitoring systems [70]. These sensors continuously monitor critical parameters such as vibration, temperature, pressure, and wear, providing a detailed view of a component's operational state. With continuous monitoring, any deviations from normal operating conditions can be detected early, allowing maintenance actions to be scheduled before failures occur. Integrating condition-based monitoring, which relies on actual condition data rather than fixed time intervals, further enhances predictive capabilities.

Another key method is the use of machine learning algorithms. These algorithms can be trained on historical data to learn patterns associated with impending failures. Supervised learning models like random forests [61], support vector machines [62], and neural networks [63] are widely used to predict component failures based on historical sensor data. More advanced techniques such as deep learning (e.g., convolutional neural networks [71] and recurrent neural networks [72]) allow predictive models to analyze raw data and identify complex relationships without the need for manual feature engineering. Unsupervised learning methods, such as anomaly detection using autoencoders [73] or k-means clustering [74], are also useful for identifying abnormal behavior that might signal early failure.

Another way to improve predictive maintenance accuracy is through remaining useful life estimation. RUL models estimate the time remaining before a component will fail based on its current condition and degradation trend. Approaches such as Weibull distribution models, linear degradation models, and particle filtering provide reliable estimates of the RUL, helping operators to schedule maintenance actions at the optimal time. Combining data-driven models with physics-based models, which simulate the physical behavior of a component under operational stresses, leads to even greater accuracy by incorporating domain-specific knowledge about how components degrade over time.

Vibration and acoustic analyses are another important method for improving accuracy. By analyzing changes in vibration patterns or acoustic emissions from components like bearings and rotating machinery, early signs of wear and imbalance can be detected. Techniques such as Fourier transforms [75] and wavelet analysis [76] enable the detection of subtle changes in vibration or sound that may indicate the onset of mechanical failure.

Cloud-based platforms for big data integration and digital twins [54] offer additional improvements in predictive maintenance accuracy. Cloud computing allows predictive

maintenance systems to process and analyze vast amounts of data in real time, while digital twins simulate the behavior of physical components in a virtual environment, enabling predictive models to run failure scenarios and optimize maintenance decisions based on real-world operating conditions. These platforms also allow for the integration of multiple data sources and analytics tools, improving the overall predictive capabilities of maintenance systems.

Statistical methods, such as time series analysis [77] and survival analysis [78], also play a key role in predictive maintenance. Time-series forecasting techniques like ARIMA (auto-regressive integrated moving average) [79] or exponential smoothing [80] help to predict future component performance based on historical data trends. Survival analysis estimates the probability that a component will continue to function without failure over time, providing critical insights into when maintenance actions should be taken. Similarly, regression analysis can be used to quantify the relationship between different operational factors and component failure rates, enabling more accurate failure predictions.

The implementation of real-time alert systems and edge computing can significantly improve predictive maintenance by reducing latency and enabling immediate responses to critical component conditions [81]. Edge computing processes data locally on devices or sensors, providing immediate insights and reducing the dependency on cloud-based processing. When anomalies are detected, real-time alerts notify maintenance teams of potential issues, allowing them to take swift action before the component fails. These alert systems can also integrate with existing decision support systems to provide recommendations for maintenance, repair, or replacement actions.

Improving predictive maintenance accuracy involves a combination of advanced data collection, sophisticated machine learning algorithms, real-time monitoring, and the integration of both data-driven and physics-based models. By using these methods, predictive maintenance systems can more accurately forecast component failures, optimize maintenance schedules, and ultimately reduce costs while maintaining high levels of safety and reliability in complex systems such as aircrafts.

5. Conclusions

This study presents a comprehensive framework for aircraft component life cycle management, introducing several significant innovations in maintenance decision-making and cost optimization. The research findings demonstrate the effectiveness of integrating genetic algorithm-based optimization with real-time health monitoring and predictive maintenance strategies.

Key findings of the case study demonstrate significant cost reductions through optimization, with mechanical components showing a 10% more reduction in total life cycle costs, avionics components achieving a 14% more cost reduction, and engine components demonstrating a 7% more decrease in total costs.

The research makes several notable contributions to the field, including the development of an integrated framework combining real-time health monitoring, reliability assessment, and economic analysis, enabling optimized maintenance decisions; the implementation of a GA-based optimization algorithm that effectively balances cost minimization with reliability requirements; the creation of a dynamic re-evaluation model that continuously assesses component health and economic feasibility; and the introduction of a comprehensive health index incorporating multiple reliability metrics (MTBF, failure rate, condition monitoring, and degradation rate).

The proposed framework addresses critical challenges in current MRO processes, including reduction in no fault found cases through improved diagnostics, enhanced component lifecycle tracking and prediction, improved feedback loops between maintenance actions and component performance, and more efficient resource allocation through optimized maintenance scheduling.

The sensitivity analysis confirmed the framework's robustness and adaptability to varying operational conditions, demonstrating its practical value for real-world implementation.

The model showed particular strength in adjusting maintenance strategies in response to changing failure rates and degradation patterns while maintaining cost-effectiveness.

Future research directions should focus on a further enhancement of predictive maintenance accuracy through advanced machine learning algorithms, the integration of emerging IoT sensor technologies for improved real-time monitoring, the development of more sophisticated digital twin implementations, and the extension of the optimization framework to handle multiple interdependent components simultaneously.

The framework's success in achieving substantial cost savings across different component types, combined with its ability to adapt to varying operational conditions, suggests strong potential for widespread adoption in the aviation industry.

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References

1. Diamanti, K.; Soutis, C. Structural Health Monitoring Techniques for Aircraft Composite Structures. *Prog. Aerosp. Sci.* **2010**, *46*, 342–352. [CrossRef]
2. Khan, S.; Phillips, P.; Jennions, I.; Hockley, C. No Fault Found Events in Maintenance Engineering Part 1: Current Trends, Implications and Organizational Practices. *Reliab. Eng. Syst. Saf.* **2014**, *123*, 183–195. [CrossRef]
3. Pelham, J. What Are the Safety Implications of No Fault Found? Available online: <https://insights.cranfield.ac.uk/blog/what-are-the-safety-implications-of-no-fault-found> (accessed on 7 October 2024).
4. Correa, L. Real Life Essentials in Five Minute Case Studies: Recommendations to Reduce NFF (No Fault Found) Rate. *Aircraft IT MRO*, July/August 2020. Available online: <https://www.aircraftit.com/articles/real-life-essentials-in-five-minute-case-studies-recommendations-to-reduce-nff-no-fault-found-rate/> (accessed on 7 October 2024).
5. The Real Cost of No Fault Found. Available online: <https://www.supportrobotics.com/our-blog/the-real-cost-of-no-fault-found> (accessed on 7 October 2024).
6. Kasturi, T. But Nothing Was Actually Broken: No Fault Found Warranty Wastage. Available online: <https://www.predii.com/post/but-nothing-was-actually-broken-no-fault-found-warranty-wastage> (accessed on 7 October 2024).
7. Smith, S.D. Minimizing Wasted Resources in No Fault Found Returns. Available online: <https://www.sdexec.com/warehousing/article/12184784/moduslink-corporation-minimizing-wasted-resources-in-no-fault-found-returns> (accessed on 7 October 2024).
8. Molent, L.; Aktepe, B. Review of Fatigue Monitoring of Agile Military Aircraft. *Fatigue Fract. Eng. Mater. Struct.* **2000**, *23*, 767–785. [CrossRef]
9. Pfingstl, S.; Steinweg, D.; Zimmermann, M.; Hornung, M. On the Potential of Extending Aircraft Service Time Using Load Monitoring. *J. Aircr.* **2022**, *59*, 377–385. [CrossRef]
10. Lee, H.; Cho, H.; Park, S. Review of the F-16 Individual Aircraft Tracking Program. *J. Aircr.* **2012**, *49*, 1398–1405. [CrossRef]
11. Jiao, R.; He, X.F.; Li, Y.H. Individual Aircraft Life Monitoring: An Engineering Approach for Fatigue Damage Evaluation. *Chin. J. Aeronaut.* **2018**, *31*, 727–739. [CrossRef]
12. Zhang, Y.; Wang, B.; Ning, Y.; Xue, H.; Lei, X. Study on Health Monitoring and Fatigue Life Prediction of Aircraft Structures. *Materials* **2022**, *15*, 8606. [CrossRef]
13. Hoffman, P.C. Fleet Management Issues and Technology Needs. *Int. J. Fatigue* **2009**, *31*, 1631–1637. [CrossRef]
14. Nicolas, M.J.; Sullivan, R.W.; Richards, W.L. Large Scale Applications Using FBG Sensors: Determination of In-Flight Loads and Shape of a Composite Aircraft Wing. *Aerospace* **2016**, *3*, 18. [CrossRef]
15. Zhang, Y.; Cao, S.; Wang, B.; Yin, Z. A Flight Parameter-Based Aircraft Structural Load Monitoring Method Using a Genetic Algorithm Enhanced Extreme Learning Machine. *Appl. Sci.* **2023**, *13*, 4018. [CrossRef]
16. Castellani, M.; Lemmens, Y.; Cooper, J.E. Parametric Reduced Order Model Approach for Rapid Dynamic Loads Prediction. *Aerosp. Sci. Technol.* **2016**, *52*, 29–40. [CrossRef]
17. Candon, M.; Esposito, M.; Fayek, H.; Levinski, O.; Koschel, S.; Joseph, N.; Carrese, R.; Marzocca, P. Advanced Multi-Input System Identification for next Generation Aircraft Loads Monitoring Using Linear Regression, Neural Networks and Deep Learning. *Mech. Syst. Signal Proc.* **2022**, *171*, 108809. [CrossRef]

18. Zhi, W.; Wang, Y.Y. A Study of the Method for Calculating Fatigue Damage of Aircraft by Using Recorded Load Factors. In Proceedings of the Congress of the International Council of the Aeronautical Sciences, Nice, France, 19 September 2010; pp. 1–7. Available online: https://www.icas.org/ICAS_ARCHIVE/ICAS2010/PAPERS/013.PDF (accessed on 5 October 2024).
19. Lee, H.; Park, S.; Kim, H. Estimation of Aircraft Structural Fatigue Life Using the Crack Severity Index Methodology. *J. Aircr.* **2010**, *47*, 1672–1678. [[CrossRef](#)]
20. White, P.; Mongru, D.; Molent, L. A Crack Growth-Based Individual Aircraft Monitoring Method Utilising a Damage Metric. *Struct. Health Monit.* **2018**, *17*, 1178–1191. [[CrossRef](#)]
21. Molent, L.; Barter, S.; Foster, W. Verification of an Individual Aircraft Fatigue Monitoring System. *Int. J. Fatigue* **2012**, *43*, 128–133. [[CrossRef](#)]
22. Kabashkin, I.; Perekrestov, V.; Tyncherov, T.; Shoshin, L.; Susanin, V. Framework for Integration of Health Monitoring Systems in Life Cycle Management for Aviation Sustainability and Cost Efficiency. *Sustainability* **2024**, *16*, 6154. [[CrossRef](#)]
23. Boller, C.; Chang, F.; Fujino, Y. *Encyclopedia of Structural Health Monitoring*; John Wiley & Sons: Hoboken, NJ, USA, 2009.
24. Corcoran, J. Rate-based structural health monitoring using permanently in-installed sensors. *Proc. R. Soc. A Math. Phys. Eng. Sci.* **2017**, *473*, 20170270.
25. Rytter, A. Vibration Based Inspection of Civil Engineering Structures. Ph.D. Thesis, Aalborg University, Aalborg, Denmark, 1993.
26. Ricci, F.; Monaco, E.; Boffa, N.D.; Maio, L.; Memmolo, V. Guided waves for structural health monitoring in composites: A review and implementation strategies. *Prog. Aerosp. Sci.* **2022**, *129*, 100790. [[CrossRef](#)]
27. Mitra, M.; Gopalakrishnan, S. Guided wave based structural health monitoring: A review. *Smart Mater. Struct.* **2016**, *25*, 053001. [[CrossRef](#)]
28. Memmolo, V.; Moll, J.; Nguyen, D.H.; Krozer, V. Interaction of guided electromagnetic waves with defects emerging in metallic plates. In Proceedings of the 2021 IEEE International Workshop on Metrology for AeroSpace, MetroAeroSpace 2021, Virtual, 23–25 June 2021; pp. 552–557.
29. Kexel, C.; Moll, J. Detecting damage in rudder stocks under load using electro-mechanical susceptance: Frequency-warping and semi-supervised approaches. *J. Intell. Mater. Syst. Struct.* **2022**, *33*, 1705–1718. [[CrossRef](#)]
30. Ooijevaar, T. Vibration Based Structural Health Monitoring of Composite Skin-Stiffener Structures. Ph.D. Thesis, University of Twente, Enschede, The Netherlands, 2014.
31. Druet, T.; Recoquillay, A.; Chapuis, B.; Moulin, E. Passive guided wave tomography for structural health monitoring. *J. Acoust. Soc. Am.* **2019**, *146*, 2395–2403. [[CrossRef](#)] [[PubMed](#)]
32. Parodi, M.; Fiaschi, C.; Memmolo, V.; Ricci, F.; Maio, L. Interaction of Guided Waves with Delamination in a Bilayered Aluminum-Composite Pressure Vessel. *J. Mater. Eng. Perform.* **2019**, *28*, 3281–3291. [[CrossRef](#)]
33. Kralovec, C.; Schagerl, M. Review of Structural Health Monitoring Methods Regarding a Multi-Sensor Approach for Damage Assessment of Metal and Composite Structures. *Sensors* **2020**, *20*, 826. [[CrossRef](#)] [[PubMed](#)]
34. Memmolo, V.; Monaco, E.; Ricci, F.; Vaselli, C.; Cimminiello, N.; Salvato, P. Structural Health Monitoring of Electromechanical Actuators in Aviation—Challenges Ahead and Case Study. *J. Nondestruct. Eval.* **2022**, *5*, 041004. [[CrossRef](#)]
35. Tschoke, K.; Mueller, I.; Memmolo, V.; Moix-Bonet, M.; Moll, J.; Lugovtsova, Y.; Golub, M.; Venkat, R.S.; Schubert, L. Feasibility of Model-Assisted Probability of Detection Principles for Structural Health Monitoring Systems based on Guided Waves for Fibre-Reinforced Composites. *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **2021**, *68*, 3156–3173. [[CrossRef](#)]
36. United States Government Accountability Office. *Aviation Safety: Status of FAA's Actions to Oversee the Safety of Composite Airplanes*; United States Government Accountability Office: Washington, DC, USA, 2011.
37. Abrate, S. *Impact on Composite Structures*; Cambridge University Press: Cambridge, UK, 1998.
38. Maio, L.; Monaco, E.; Ricci, F.; Lecce, L. Simulation of low velocity impact on composite laminates with progressive failure analysis. *Compos. Struct.* **2013**, *103*, 75–85. [[CrossRef](#)]
39. Department of Defense. *Composite Materials Handbook*; MIL-HDBK 17-3F; U.S. Department of Defense: Washington, DC, USA, 2002.
40. U.S. Department of Transportation. *Composite Aircraft Structures*; AC No. 20-107B; U.S. Department of Transportation—Federal Aviation Administration: Washington, DC, USA, 2009.
41. U.S. Department of Transportation. *Damage Tolerance and Fatigue Evaluation of Structures*; AC No. 25,571-1D; U.S. Department of Transportation—Federal Aviation Administration: Washington, DC, USA, 2011.
42. Fioriti, M.; Pavan, G.; Corpino, S.; Fusaro, R. Impacts of a prognostics and health management system on aircraft fleet operating cost during conceptual design phase by using parametric estimation. In Proceedings of the 5th CEAS Air & Space Conference, Delft, The Netherlands, 7–11 September 2015; Available online: <http://hdl.handle.net/11583/2627572> (accessed on 15 May 2021).
43. Leao, B.P.; Fitzgibbon, K.T.; Puttini, L.C.; de Melo, G.P. Cost-benefit analysis methodology for PHM applied to legacy commercial aircraft. In Proceedings of the 2008 IEEE Aerospace Conference, Big Sky, MT, USA, 1–8 March 2008; pp. 1–14.
44. Feldman, K.; Jazouli, T.; Sandborn, P.A. A methodology for determining the return on investment associated with prognostics and health management. *IEEE Trans. Reliab.* **2009**, *58*, 305–316. [[CrossRef](#)]
45. Dong, T.; Kim, N. Cost-effectiveness of structural health monitoring in fuselage maintenance of the civil aviation industry. *Aerospace* **2018**, *5*, 87. [[CrossRef](#)]
46. Ahmadi, A.; Söderholm, P.; Kumar, U. On Aircraft Scheduled Maintenance Program Development. *J. Qual. Maint. Eng.* **2010**, *16*, 229–255. [[CrossRef](#)]

47. Airline/Manufacturer Maintenance Program Development Document MSG-3 Revision 2. Available online: <https://pdfcoffee.com/msg-3-rev-2-pdf-free.html> (accessed on 5 October 2024).
48. Walthall, R.; Rajamani, R. The Role of PHM at Commercial Airlines. In *Prognostics and Health Management of Electronics*; John Wiley & Sons, Ltd.: Hoboken, NJ, USA, 2018; pp. 503–534. ISBN 978-1-119-51532-6.
49. Millwater, H.R.; Wieland, D.H. Probabilistic Sensitivity-Based Ranking of Damage Tolerance Analysis Elements. *J. Aircr.* **2010**, *47*, 161–171. [\[CrossRef\]](#)
50. Skinn, D.A.; Gallagher, J.P.; Berens, A.P.; Huber, P.D.; Smith, J. *Damage Tolerant Design Handbook*; Defense Technical Information Center: Fort Belvoir, VA, USA, 1994; p. 745.
51. Newcamp, J.; Verhagen, W.J.C.; Curran, R. Correlation of Mission Type to Cyclic Loading as a Basis for Agile Military Aircraft Asset Management. *Aerosp. Sci. Technol.* **2016**, *55*, 111–119. [\[CrossRef\]](#)
52. Grooteman, F.P. *Probabilistic Fail-Safe Structural Risk Analyses*; Netherlands Aerospace Centre NLR: Amsterdam, The Netherlands, 2020; p. 24. Available online: <https://reports.nlr.nl/server/api/core/bitstreams/95fa6b86-bb2c-44dc-9fd5-36aa8c400e30/content> (accessed on 5 October 2024).
53. Souanef, T.; Al-Rubaye, S.; Tsourdos, A.; Ayo, S.; Panagiotakopoulos, D. Digital Twin Development for the Airspace of the Future. *Drones* **2023**, *7*, 484. [\[CrossRef\]](#)
54. Kabashkin, I. Digital Twin Framework for Aircraft Lifecycle Management Based on Data-Driven Models. *Mathematics* **2024**, *12*, 2979. [\[CrossRef\]](#)
55. Laukotka, F.N.; Krause, D. Supporting Digital Twins for the Retrofit in Aviation by a Model-Driven Data Handling. *Systems* **2023**, *11*, 142. [\[CrossRef\]](#)
56. Kabashkin, I. NFT-Based Framework for Digital Twin Management in Aviation Component Lifecycle Tracking. *Algorithms* **2024**, *17*, 494. [\[CrossRef\]](#)
57. Tian, X.-Y.; Shi, X.; Peng, C.; Yi, X.-J. A Reliability Growth Process Model with Time-Varying Covariates and Its Application. *Mathematics* **2021**, *9*, 905. [\[CrossRef\]](#)
58. Lee, D.; Chang, I.; Pham, H. Study of a New Software Reliability Growth Model under Uncertain Operating Environments and Dependent Failures. *Mathematics* **2023**, *11*, 3810. [\[CrossRef\]](#)
59. Żyluk, A.; Kuźma, K.; Grzesik, N.; Zieja, M.; Tomaszewska, J. Fuzzy Logic in Aircraft Onboard Systems Reliability Evaluation—A New Approach. *Sensors* **2021**, *21*, 7913. [\[CrossRef\]](#)
60. Hu, X.; Ren, H. Reliability Estimation of Inverse Weibull Distribution Based on Intuitionistic Fuzzy Lifetime Data. *Axioms* **2023**, *12*, 838. [\[CrossRef\]](#)
61. Wang, X.; Wang, R.; Wei, S.; Xu, S. Application of Random Forest Method Based on Sensitivity Parameter Analysis in Height Inversion in Changbai Mountain Forest Area. *Forests* **2024**, *15*, 1161. [\[CrossRef\]](#)
62. Guido, R.; Ferrisi, S.; Lofaro, D.; Conforti, D. An Overview on the Advancements of Support Vector Machine Models in Healthcare Applications: A Review. *Information* **2024**, *15*, 235. [\[CrossRef\]](#)
63. Taherdoost, H. Deep Learning and Neural Networks: Decision-Making Implications. *Symmetry* **2023**, *15*, 1723. [\[CrossRef\]](#)
64. Poza, J.; Moreno, V.; Fraga, A.; Álvarez-Rodríguez, J.M. Genetic Algorithms: A Practical Approach to Generate Textual Patterns for Requirements Authoring. *Appl. Sci.* **2021**, *11*, 11378. [\[CrossRef\]](#)
65. Cui, X.; Li, X.; Zhao, Z.; Yu, J.; Liu, D. Research on Classification Maintenance Strategy for More Electric Aircraft Actuation Systems Based on Importance Measure. *Symmetry* **2024**, *16*, 1127. [\[CrossRef\]](#)
66. Oujana, S.; Amodeo, L.; Yalaoui, F.; Brodard, D. Mixed-Integer Linear Programming, Constraint Programming and a Novel Dedicated Heuristic for Production Scheduling in a Packaging Plant. *Appl. Sci.* **2023**, *13*, 6003. [\[CrossRef\]](#)
67. Barraza-Contreras, J.M.; Piña-Monarez, M.R.; Molina, A. Fatigue-Life Prediction of Mechanical Element by Using the Weibull Distribution. *Appl. Sci.* **2020**, *10*, 6384. [\[CrossRef\]](#)
68. Raza, A.; Ulansky, V. Through-Life Maintenance Cost of Digital Avionics. *Appl. Sci.* **2021**, *11*, 715. [\[CrossRef\]](#)
69. Qi, Y.; Jiang, B.; Lei, W.; Zhang, Y.; Yu, W. Reliability Analysis of Normal, Lognormal, and Weibull Distributions on Mechanical Behavior of Wood Scrimber. *Forests* **2024**, *15*, 1674. [\[CrossRef\]](#)
70. Kabashkin, I.; Perekestov, V. Ecosystem of Aviation Maintenance: Transition from Aircraft Health Monitoring to Health Management Based on IoT and AI Synergy. *Appl. Sci.* **2024**, *14*, 4394. [\[CrossRef\]](#)
71. Krichen, M. Convolutional Neural Networks: A Survey. *Computers* **2023**, *12*, 151. [\[CrossRef\]](#)
72. Mienye, I.D.; Swart, T.G.; Obaido, G. Recurrent Neural Networks: A Comprehensive Review of Architectures, Variants, and Applications. *Information* **2024**, *15*, 517. [\[CrossRef\]](#)
73. Givnan, S.; Chalmers, C.; Fergus, P.; Ortega-Martorell, S.; Whalley, T. Anomaly Detection Using Autoencoder Reconstruction upon Industrial Motors. *Sensors* **2022**, *22*, 3166. [\[CrossRef\]](#) [\[PubMed\]](#)
74. Ahmed, M.; Seraj, R.; Islam, S.M.S. The *k*-means Algorithm: A Comprehensive Survey and Performance Evaluation. *Electronics* **2020**, *9*, 1295. [\[CrossRef\]](#)
75. Bahri, M.; Abdul Karim, S.A. Fractional Fourier Transform: Main Properties and Inequalities. *Mathematics* **2023**, *11*, 1234. [\[CrossRef\]](#)
76. Nguyen, T.-D.; Nguyen, P.-D. Improvements in the Wavelet Transform and Its Variations: Concepts and Applications in Diagnosing Gearbox in Non-Stationary Conditions. *Appl. Sci.* **2024**, *14*, 4642. [\[CrossRef\]](#)

77. Jasra, S.K.; Valentino, G.; Muscat, A.; Camilleri, R. Hybrid Machine Learning–Statistical Method for Anomaly Detection in Flight Data. *Appl. Sci.* **2022**, *12*, 10261. [[CrossRef](#)]
78. Papathanasiou, D.; Demertzis, K.; Tziritas, N. Machine Failure Prediction Using Survival Analysis. *Future Internet* **2023**, *15*, 153. [[CrossRef](#)]
79. Wang, C.-C.; Chien, C.-H.; Trappey, A.J.C. On the Application of ARIMA and LSTM to Predict Order Demand Based on Short Lead Time and On-Time Delivery Requirements. *Processes* **2021**, *9*, 1157. [[CrossRef](#)]
80. Mahajan, S.; Chen, L.-J.; Tsai, T.-C. Short-Term PM2.5 Forecasting Using Exponential Smoothing Method: A Comparative Analysis. *Sensors* **2018**, *18*, 3223. [[CrossRef](#)]
81. Kabashkin, I. Unified Aviation Maintenance Ecosystem on the Basis of 6G Technology. *Electronics* **2024**, *13*, 3824. [[CrossRef](#)]

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